

1. Rectangular weir
 - a. Suppressed rectangular weir

$$Q = 0.0184 L H^{\frac{3}{2}}$$

- b. Contracted rectangular weir

$$Q = 0.0184 (L - 0.2H) H^{\frac{3}{2}}$$

Where,

Q = Discharge (litres/sec.)

L = Length of crest (cm)

H = Head over the crest (cm)

2. Trapezoidal or cipolletti weir

$$Q = 0.0186 L H^{\frac{3}{2}}$$

Where,

Q = Discharge (litres/sec.)

L = Length of crest (cm)

H = Head over the crest (cm)

3. 90° V-notch weir

$$Q = 0.0138 H^{\frac{5}{2}}$$

Where,

Q = Discharge (litres/sec.)

H = Head over the crest (cm)

4. Orifices

The discharge through an orifice is calculated by the formula

$$Q = 0.61 \times 10^{-3} A_o \sqrt{2gH}$$

Where,

Q = Discharge through orifice, litres /sec.

a = Area of cross section of the orifice, cm²

g = Acceleration due to gravity, cm /sec² (981 cm/sec²)

H = Depth of water over the centre of the orifice, cm.

5. Syphons

These are generally used to irrigate the field from canals and channels. Syphons are the pipes usually made of rubber or plastic or aluminium. Following formula can be used to measure the discharge rate of syphons :

$$Q = 0.61 \times 10^{-3} A_{\text{water@syphon}} \sqrt{2gh}$$

Where,

Q = Discharge rate of water (lit. /sec.)

g = Acceleration due to gravity (981 cm/ sec.²)

h = Height of water (cm)

A = Cross sectional area of the water in the syphon (cm²).

- 6.

PART 2 SOLVED PROBLEMS

1. The peak runoff in a 40-ha watershed corresponds to a rainfall intensity of 20mm/hr. and a runoff coefficient of 0.8 is?

Given:

Area = 40ha

Rainfall intensity (I) = 20mm/hr.

Runoff Coefficient (C) = 0.8

Solution:

$$Q = CIA = (0.8) \left(\frac{20\text{mm}}{\text{hr}} \right) (40\text{ha}) = 1.8\text{m}^3/\text{sec}$$

2. A floodway has been designed for a discharge with a return period of 75 years. The risk that the structure will experience the design flood one or more times in 20 years is

Solution:

$$P = \left(1 - \left(1 - \frac{1}{T} \right)^n \right) \times 100 = \left(1 - \left(1 - \frac{1}{75} \right)^{20} \right) \times 100 = 23.5\%$$

3. A soil sample was collected using a 3 inch diameter 10cm long core sampler. If the wet mass of the soil is 657g and the dry mass basis moisture content is 20%, the bulk density is?

Solution:

$$\text{Dry mass} = 657 \times 0.20 = 525.60\text{g}$$

$$\text{BD} = \frac{M_d}{V_T} = \frac{525.6\text{g}}{\frac{\pi(3 \times 2.54\text{cm})^2(10\text{cm})}{4}} = \frac{1.20\text{g}}{\text{cm}^3}$$

4. If the bulk density of the soil is 1.10g/cc, the soil porosity is?

Solution:

$$\eta = 1 - \frac{\frac{1.10\text{g}}{\text{cm}^3}}{\frac{2.65\text{g}}{\text{cm}^3}} = 58\%$$

Note: 2.65g/cc is the particle density average value.

5. A 0.25ha cornfield was irrigated for 5hrs at a discharge of 32lps. If the average depth of water stored in the root-zone as a result of irrigation was observed to be 160mm, the application efficiency is?

Solution:

$$\text{Eff}_a = \frac{(0.25\text{ha})(10,000\text{m}^2)(0.16\text{m})}{(32\text{lps})(0.001\text{m}^3)(3600\text{sec})} = 70\%$$

6. The gross application depth is 78mm to irrigate a crop with a crop coefficient of 0.9. The reference crop evapotranspiration is 6.5mm/day. If the application efficiency is 60% and assuming rainless days, irrigation should be done every

Solution:

$$\text{II} = \frac{d_n}{\text{ET}_{\text{crop}}} = \frac{d_g E_a}{K_c \text{ET}_o} = \frac{(78\text{mm})(0.6)}{(0.9) \left(\frac{6.5\text{mm}}{\text{day}} \right)} = 8\text{days}$$

7. If a 300mm depth of soil, with a porosity of 48%, residual moisture content of 20% dry mass basis and bulk density is 1.1g/cc, is to be soaked for 5 days, the land soaking requirement should be

Solution:

$$\text{LSR} = \frac{(\eta - \text{RMC}_s)(d_{rz})}{\Delta t} = \frac{(0.48 - (0.2)(1.1))(300\text{mm})}{5\text{days}} = \frac{15.6\text{mm}}{\text{day}}$$

8. The crop water requirement is 15mm/day and the effective rainfall is 2mm/day. If the application efficiency is 75% and the conveyance efficiency is 80%, the diversion water requirement in lps/ha is

Solution:

$$DWR = \frac{\text{Field water requirement (FWR)}}{\text{Conveyance Efficiency}} = \frac{\text{Field irrigation requirement (FIR)}}{\text{Application Efficiency} \times \text{Conveyance Efficiency}}$$

$$DWR = \frac{FIR}{\text{Eff}_a \text{Eff}_c} = \frac{\text{Crop Water Requirement} - \text{Effective Rainfall}}{\text{Eff}_a \text{Eff}_c}$$

$$DWR = \frac{\frac{15\text{mm}}{\text{day}} - \frac{2\text{mm}}{\text{day}}}{(0.75)(0.8)} = \frac{21.67\text{mm}}{\text{day}} \left(\frac{\frac{1\text{lps}}{\text{ha}}}{\frac{8.64\text{mm}}{\text{day}}} \right) = \frac{2.50\text{lps}}{\text{ha}}$$

Note:

$$FWR = FIR/\text{Eff}_a$$

$$FIR = CWR - \text{ERF}$$

9. Given the following data for peak period of upland rice.

$$ET_o = 6\text{mm per day}$$

$$\text{Percolation} = 3\text{mm per day}$$

$$\text{ERF} = 12\text{mm in 10days}$$

$$\text{Eff}_a = 70\%$$

$$K_c = 1.17$$

$$\text{Eff}_c = 80\%$$

Turnout No.	Area (ha)
1	12
2	15
3	18

Compute the Farm turnout requirement in lps

Solution:

$$FWR \text{ or } FTR = \frac{FIR}{\text{Eff}_a} = \frac{CWR - \text{ERF}}{\text{Eff}_a} = \frac{ET_{\text{crop}} + P - \text{ERF}}{\text{Eff}_a}$$

$$FWR \text{ or } FTR = \frac{K_c ET_o + P - \text{ERF}}{\text{Eff}_a} = \frac{(1.17) \left(\frac{6\text{mm}}{\text{day}} \right) + \left(\frac{3\text{mm}}{\text{day}} \right) - \left(\frac{1.2\text{mm}}{\text{day}} \right)}{0.70}$$

$$FWR \text{ or } FTR = \frac{12.6\text{mm}}{\text{day}} \left(\frac{\frac{1\text{lps}}{\text{ha}}}{\frac{8.64\text{mm}}{\text{day}}} \right) = \frac{1.46\text{lps}}{\text{ha}} (45\text{ha}) = 66\text{lps}$$

10. The demand flow during the peak period in problem #9 is

Solution:

$$Q_{\text{demand}} = DWR(\text{Adjusted Irrigable area}) = \frac{FWR(\text{IA}_{\text{adj}})}{\text{Eff}_c} = \frac{66\text{lps}}{0.8} = 82\text{lps}$$

11. The farm water requirement is 2lps/ha, application efficiency is 75% and the conveyance efficiency is 80%. If the dependable flow is 2.5m³/s, the irrigable area is

Solution:

$$IA = \frac{Q_{\text{dep}}}{\text{DWR}} = \frac{Q_{\text{dep}}}{\frac{\text{FWR}}{\text{Eff}_c}} = \frac{\left(\frac{2.5\text{m}^3}{\text{s}}\right)(0.8)}{\frac{2\text{lbs}}{\frac{\text{ha}}{1,000\text{lit}}}} = 1000\text{ha}$$

12. Three tanks are installed side by side in a field to measure the evapotranspiration of rice. Tank A is bottomless and is cropped. Tank B is bottomless but is uncropped. Tank C has bottom and is uncropped. If the water losses after 2 rainless days are as follows

Tank A – 15mm

Tank B – 9mm

Tank C – 5mm

The actual evapotranspiration is

$$ET_a = \frac{A - B + C}{2} = \frac{15 - 9 + 5}{2} = \frac{5.5\text{mm}}{\text{day}}$$

13. A soil sample was obtained using a 30cm diameter can with corresponding height of 50cm.

- a. Compute the total volume of the sample.

Solution:

$$V_t = \frac{\pi D^2 L}{4} = \frac{\pi(30\text{cm})^2(50\text{cm})}{4} = 35,343\text{cm}^3$$

- b. If the void volume is equal to 30,150cm³, find its porosity

Solution:

$$\eta = \frac{V_v}{V_t} = \frac{30,150\text{cm}^3}{35,343\text{cm}^3} = 85.3\%$$

- c. Find the dry soil volume

Solution:

$$V_d = 35,343\text{cm}^3 - 30,150\text{cm}^3 = 5,192\text{cm}^3$$

14. The recent monsoon rain have posted a nearly uniform rainfall intensity of 30mm/hr. over a 1500ha watershed area in central Luzon. If the runoff coefficient is 0.15, what is the runoff in m³/sec?

Solution:

$$Q = CIA = (0.15) \left(\frac{0.030\text{m}}{\text{hr}}\right) (1500\text{ha}) \left(\frac{10,000\text{m}^2}{3,600\text{s}}\right) = \frac{18.75\text{m}^3}{\text{s}}$$

15. An artesian well has a diameter of 10cm under steady state condition. What is the expected discharge (lps) if the drawdown is 2.5m and the radius of influence is 120m? the aquifer's hydraulic conductivity and thickness are 75m/day and 4m, respectively.

Solution:

$$Q = \frac{2\pi k t (h_i - h_w)}{\ln\left(\frac{r_i}{r_w}\right)} = \frac{2\pi \left(\frac{75\text{m}}{\text{day}}\right) (4\text{m}) (2.5\text{m})}{\ln\left(\frac{120\text{m}}{0.05\text{m}}\right)} = \frac{605.45\text{m}^3}{\text{day}} \left(\frac{1\text{day}}{86400\text{s}}\right) \left(\frac{1000\text{lit}}{1\text{m}^3}\right) = 7.01\text{lps}$$

16. A 21.6mm/day water requirement is equivalent to

Solution:

$$\frac{21.6\text{mm}}{\text{day}} \left(\frac{\frac{1\text{lbs}}{\text{ha}}}{\frac{8.64\text{mm}}{\text{day}}}\right) = 2.5\text{lps/ha}$$

17. An irrigation engineer, in his study, able to establish that the irrigation requirement of crop xyz is 6.5mm/day. If the root-zone depth was estimated at 2 inch on the average, what should be the design discharge of a canal to be able to deliver the 5-days requirement of a 10-ha farm in 24hrs?

Solution:

$$Q = \frac{Ad_g}{t} = \frac{(10 \times 10,000 \text{ m}^2) \left(\frac{6.5 \text{ mm}}{\text{day}} \right) (5 \text{ days})}{(1,000 \text{ mm})(24 \text{ hrs})(3600 \text{ sec})} = \frac{0.03762 \text{ m}^3}{\text{sec}} (1,000 \text{ lit}) = 37.62 \text{ lps}$$

18. An irrigation engineer is to irrigate a cropland in furrows of 100m x 100m area. It was found out that the inflow rate is 0.6 lps per 100m. If he decides to have a furrow spacing of 50cm and application depth of 100mm, determine:

- a. The time of irrigating each furrow.

Solution:

$$t = \frac{Ad_g}{Q} = \frac{(10,000 \text{ m}^2)(0.1 \text{ m})}{\left(\frac{0.5 \text{ lit}}{\text{s}} \right) \left(\frac{3600 \text{ s}}{\text{hr}} \right) \left(\frac{1 \text{ m}^3}{1000 \text{ lit}} \right)} = 555.56 \text{ hrs or } 556 \text{ hrs}$$

- b. The time to irrigate each furrow

Solution:

$$t = \frac{555.56 \text{ hrs}}{200 \text{ furrow}} = 2.78 \text{ hrs}$$

Note: No. of furrows

$$X = 100 \text{ m} / 0.5 \text{ m} = 200 \text{ furrows}$$

19. The soil moisture content of corn field is 20% by weight prior to irrigation. Field capacity is 28% by weight. Determine the net application depth if the apparent specific gravity is 1.40 and the rooting depth is 0.8m, management allowable depletion is 50%

Solution:

$$a. d_{\text{net}} = \text{MAD}(\text{FC} - \text{SMC})A_s D_{\text{rz}} = (0.5)(0.28 - 0.20)(1.40)(0.8 \text{ m}) = 0.0448 \text{ m or } 4.48 \text{ cm}$$

Note: PWP was replaced with soil moisture content (SMC) because the soil is already prior to irrigation. Means the cornfield has reached the PWP

- b. If Eff_a is 60%, determine the gross application depth.

Solution

$$d_g = \frac{d_n}{\text{Eff}_a} = \frac{4.48 \text{ cm}}{0.6} = 7.47 \text{ cm}$$

- c. Calculate the minimum time if the furrow intake rate is 8mm/hr.

Solution:

$$t = \frac{d_g}{I} = \frac{7.47 \text{ cm}}{\frac{8 \text{ mm}}{\text{hr}}} = 9.34 \text{ hr}$$

- d. What should be the stream size if the area to be irrigated is 3-ha?

Solution:

$$Q = \frac{Ad_g}{t} = \frac{(3 \times 10,000 \text{ m}^2)(0.0747 \text{ m})(1000 \text{ lit})}{(9.34 \text{ hr})(3600 \text{ sec})} = 66.65 \text{ lps}$$

- e. When was the last irrigation applied if the consumptive use of the corn crop has been 5mm/day?

Solution:

$$\text{Last irrig. applied} = \frac{d_n}{\text{C. U.}} = \frac{44.8 \text{ mm}}{\frac{5 \text{ mm}}{\text{day}}} = 8.96 \text{ days or } 9 \text{ days}$$

20. A concrete lined canal is to convey water over a distance of 250 m from the well to a tomato farm. Drop in elevation is 0.5 m over the 250 m canal length and roughness coefficient, n , is 0.018.

- a. Estimate the bottom width of a rectangular canal with a depth of flowing water of 0.25 m if the hydraulic radius is equal to the area of flow.

Solution:

Since $A = R$, & $P = 4y$ for rectangle,

$$P = b + 2y$$

$$4y = b + 2y$$

$$2y = b$$

$$b = 2(0.25) = 0.5$$

- b. The velocity of the canal

Solution:

$$\text{Since } R = A = 2y^2 = 2(0.25\text{m})^2 = 0.125$$

$$V = \frac{1}{n} R^{\frac{2}{3}} S^{\frac{1}{2}} = \frac{1}{0.018} (0.125\text{m})^{\frac{2}{3}} \left(\frac{0.5\text{m}}{250\text{m}} \right)^{\frac{1}{2}} = \frac{0.62\text{m}}{\text{s}}$$

21. The volume of the soil sample during sampling was 85 cc and weight after drying in the area was 130g.

- a. The volume weight of the soil is

Solution:

Volume weight is the same as bulk density

$$BD = \frac{130\text{g}}{85\text{cm}^3} = \frac{1.53\text{g}}{\text{cm}^3}$$

- b. The percentage of moisture by weight in (a) if the weight of the sample before drying was 145 g.

Solution:

$$\%_w = \frac{145 - 130}{130} = 11.54\%$$

- c. The amount of water in the soil layer of thickness 10 cm is

Solution:

$$d_w = (\%_w)(A_s)(d_{rz}) = (0.1154)(1.53)(10\text{cm}) = 1.77\text{cm}$$

22. The discharge of first emitter in a dropper line is 12 lph. What is the allowable minimum discharge in the last emitter?

Solution:

$$Q_{\text{last}} = Q_{1\text{st}} - 0.2Q_{1\text{st}} = 12\text{lph} - 0.2(12\text{lph}) = 9.6\text{lph}$$

23. What is the depth of flowing water in a trapezoidal canal with a side slope of 1:1 and base of 1.0 m if the stream velocity is 1m/s and the average discharge is $6\text{m}^3/\text{s}$.

Solution:

Area of trapezoidal canal is

$$A = bd + zd^2; Q = AV; A = Q/V = 6\text{m}^3 / 1\text{m/s} = 6\text{m}^2$$

$$A = bd + zd^2$$

$$6\text{m}^2 = d + d^2; d = 2\text{m}$$

24. The soil has a field capacity and permanent wilting point of 27% and 13%, respectively. What is the readily available moisture in cm of water per cm of soil if the bulk density is 1.25 g/cc?

Give:

$$FC = 27\%$$

$$PWP = 13\%$$

$$BD = 1.25\text{g/cc}$$

What is the RAM?

Solution:

Available water content constant is 75% and a unit depth of 1m

$$\text{RAM} = (0.75)(\text{FC} - \text{PWP})A_s * d_{rz} = (0.75)(0.27 - 0.13)(1.25)(1\text{m}) = 13.125\text{cm}$$

25. The hydraulic radius of a pipe with a diameter of 12mm is

Solution:

$$R = d/4 = 12\text{mm}/4 = 3\text{mm}$$

26. What discharge can be expected from an unconfined well 200 mm in diameter if the drawdown is 5 m in an aquifer saturated to a depth of 18 m? Assume the permeability and the radius of influence to be 20 m/day and 150 m, respectively.

Given:

Well diameter = 200mm

Drawdown = 5m

Aquifer saturation depth = 18m

Permeability = 20m/day = hydraulic conductivity

Radius of influence = 150m

Solution:

$$\text{Drawdown} = h_i - h_w; h_w = h_i - \text{Drawdown} = 18\text{m} - 5\text{m} = 13\text{m}$$

$$Q = \frac{\pi k((h_i)^2 - (h_w)^2)}{\ln\left(\frac{r_i}{r_w}\right)} = \frac{\pi\left(\frac{20\text{m}}{\text{day}}\right)((18\text{m})^2 - (13\text{m})^2)}{\ln\left(\frac{150\text{m}}{0.1\text{m}}\right)} = \frac{1331.70\text{m}^3}{\text{day}} \text{ or } 15.41\text{ps}$$

27. A developer decided to establish a golf driving range having a 122m x 183m area. If they decide to start irrigating the area using sprinklers under the following conditions:

Length of lateral	= 122 m
Spacing	= 18.3 m
Sprinkler spacing	= 12.2 m
Soil water holding capacity	= 120 mm/m
Average rooting depth	= 1.0 m
Evapotranspiration	= 6 mm/day
Readily available moisture	= 45%
Irrigation period	= 12 hrs.
Application efficiency	= 70%

- a. What would be the application rate in mm/hr.?

Solution:

$$i = \frac{d_g}{t} = \frac{d_n}{\text{Eff}_a * t} = \frac{\text{RAM} * \text{WHC} * d_{rz}}{\text{Eff}_a * t} = \frac{(0.45)\left(\frac{120\text{mm}}{\text{m}}\right)(1\text{m})}{(0.70)(12\text{hrs})} = \frac{6.43\text{mm}}{\text{hr}}$$

- b. What is the irrigation interval in days?

Solution:

$$\text{II} = \frac{d_n}{\text{ET}_{\text{crop}}} = \frac{\text{RAM} * \text{WHC} * d_{rz}}{\text{ET}_{\text{crop}}} = \frac{(0.45)\left(\frac{120\text{mm}}{\text{m}}\right)(1\text{m})}{\frac{6\text{mm}}{\text{day}}} = 9\text{days}$$

- c. What is the sprinklers discharge in lps?

Solution:

$$d_g = \frac{d_n}{\text{Eff}_a} = \frac{\text{RAM} * \text{WHC} * d_{rz}}{\text{Eff}_a} = \frac{(0.45)\left(\frac{120\text{mm}}{\text{m}}\right)(1\text{m})}{0.7} = 77.14\text{mm}$$

$$Q = \frac{Ad_g}{t} = \frac{(18.3\text{m} \times 12.2\text{m})(0.07714\text{mm})}{12\text{hrs} \left(\frac{3600\text{s}}{\text{hr}} \right)} = 0.4\text{lbs}$$

- d. What is the system capacity in lps?

Solution:

$$Q = \frac{Ad_g}{t} = \frac{(183\text{m} \times 122\text{m})(0.07714\text{m})}{(12\text{hrs})(3600\text{s})} = 39.87\text{lps}$$

28. If the rectangular canal has a base of 0.65 m, what will be the depth of flow when the hydraulic radius is equal to the area of flow?

Solution:

If $R = A$ and $A = by$ and $P = b + 2y$

$$R = \frac{by}{b + 2y}$$

$$by = \frac{by}{b + 2y}$$

$$1 = \frac{1}{b + 2y}; b + 2y = 1$$

$$y = \frac{1 - b}{2} = \frac{1 - 0.65}{2} = 0.175\text{m}$$

29. An undisturbed soil sample 100 cc was gathered from a field 2 days after heavy rains when the soil moisture was expected to be at field capacity. The weight of the fresh sample was 176 g but was reduced to 142 g after oven drying. A second soil sample with a fresh weight and oven dried weight of 130 g and 116 g, respectively, was gathered 3 weeks later when the crop was observed to have a permanently wilted.

Given:

- a. Determine the bulk density in g/cc.

Solution:

$$BD = \frac{142\text{g}}{100\text{cm}^3} = \frac{1.42\text{g}}{\text{cm}^3}$$

- b. Determine the apparent specific gravity

$$A_s = 1.42$$

- c. Determine the field capacity (dry basis).

Solution:

$$FC = \frac{FW - OD}{OD} = \frac{176\text{g} - 142\text{g}}{142\text{g}} = 23.94\%$$

- d. Determine the wilting point (dry basis).

Solution:

$$PWP = \frac{FW - OD}{OD} = \frac{130\text{g} - 116\text{g}}{116\text{g}} = 12.07\%$$

30. A crop has a depth of 0.6 m. F.C. and W.P. are 16% and 8%, respectively. Bulk density is 1.25 g/cc. The desired allowable soil moisture deficit is 60% of the allowable moisture and the average consumptive use is 6 mm/day.

- a. The depth of water to be stored in the root-zone per irrigation is

Solution:

$$d_n = MAD(FC - PWP)A_s * d_{rz} = (0.6)(0.16 - 0.08)(1.25)(0.6\text{m}) = 0.036\text{m}$$

- b. The irrigation interval is:

Solution:

$$II = \frac{d_n}{ET_{\text{crop}}} = \frac{36\text{mm}}{\frac{6\text{mm}}{\text{day}}} = 6\text{days}$$

31. The net peak crop water requirement for a scheme is 6.0 mm/day. The available moisture for the clay loam is 130mm/m and depletion is allowed up to around 46%. The root zone depth is 0.70 m. After how many days should irrigation take place to replenish the soil moisture?

Solution:

$$II = \frac{d_n}{ET_{\text{crop}}} = \frac{RAM * WHC * d_{rz}}{ET_{\text{crop}}} = \frac{(0.46) \left(\frac{130\text{mm}}{\text{m}} \right) (0.70\text{m})}{\frac{6\text{mm}}{\text{day}}} = 6.977\text{days or } 7\text{days}$$

32. At Nabusenga surface irrigation schemes, it is assumed that conveyance efficiency and field canal efficiency both are 90% for concrete-lined canals continuous flow and that application efficiency is 50%. What is the overall irrigation efficiency?

Solution:

$$E_t = (0.9)(0.9)(0.5) = 40.5\%$$

33. What is the water depth for a trapezoidal canal with the following known parameters:

$$Q = 0.09\text{m}^3/\text{sec}$$

$$S = 0.001$$

$$b = y$$

$$n = 55$$

$$\text{Side slope } (z) = 1$$

$$\text{Allowable Velocity} = <0.75\text{m/sec}$$

Solution:

$$A = by + zy^2 = y^2 + y^2 = 2y^2$$

$$P = b + 2y\sqrt{1 + z^2} = y + 2y\sqrt{2} = 3.838y$$

$$R = \frac{A}{P} = \frac{2y^2}{3.838y} = 0.52y$$

$$Q = AR^{\frac{2}{3}}S^{\frac{1}{2}}n$$

$$\frac{0.09\text{m}^3}{\text{s}} = (2y^2)(0.52y)^{\frac{2}{3}}(0.001)^{\frac{1}{2}}(55) = 2.25y^{2.67}$$

$$y = 0.30\text{m}$$

34. A discharge of 1.25 m³/sec has to be abstracted from a river, into an open conveyance canal. The base flow water level of the river is 125.35 m. The design water level in the canal is 124.90 m and the water depth is 0.60 m. The weir coefficient is 1.60. The width of the intake is 1.50 m and the length of the weir is 0.50 m. What will be the sill level?

Given:

$$Q = 1.25\text{m}^3/\text{sec}$$

$$\text{Weir coefficient} = 1.60$$

$$\text{Difference between river water level and canal design water level} = 125.35 - 124.90 = 0.45\text{m}$$

$$\text{Width of the intake} = 1.50\text{m}$$

Required: Difference between canal design water level and sill level of the intake (h_d)

Solution:

$$Q = C * B(h + h_d)^{\frac{3}{2}} = (1.60) * (1.50)(0.45\text{m} + h_d)^{\frac{3}{2}}$$

$$h_d = 0.20\text{m}$$

Thus the sill level should be at an elevation of $124.90 - 0.20 = 124.70\text{ m}$

35. Suppose a measurement of the air gave the temperature (T) to be 80 °F and the relative humidity (RH) to be 60 percent.

- a. Calculation of saturated vapor pressure e^o

Solution:

$$e^o = \left(\frac{164.8 + T}{157} \right)^8 = \left(\frac{164.8 + 80}{157} \right)^8 = 34.94 \text{ mb}$$

- b. Calculation of the actual vapor pressure e

Solution:

$$RH = \left(\frac{e}{e^o} \right) 100; e = 20.9 \text{ mb}$$

- c. Calculation of the dew point temperature T_d

Solution:

$$T_d = 157(e)^{0.125} - 164.8 = 157(20.9 \text{ mb})^{0.125} - 164.8 = 64.8^\circ \text{F}$$

36. In an attempt to the groundwater flow of a certain farm, two parallel canals have to be constructed. It was observed that the impervious layer exist at a depth of 1.9 m below the soil surface and the soil has a k value of 1.2 m/day. If the water table has to be maintained at a depth of 0.8 m below the soil surface, what will be the spacing of the drain when the drain level is 1.5 m below the soil. Assume a steady recharge of 2 mm/day.

Solution:

$$L^2 = \frac{4K(H + D)(H - D)}{R} = \frac{(4Kh)(2D + h)}{R} = \frac{4 \left(\frac{1.2 \text{ m}}{\text{day}} \right) (0.7)(2(0.4) + 0.7)}{\frac{0.002 \text{ mm}}{\text{day}}} = 2520 \text{ m}^2$$

$$L = 50.20 \text{ m}$$

37. You are asked to determine the size of pump to irrigate a 100m x 100m cropland under the following conditions:

- Friction head loss along main and lateral, $H_m = 18.46$ ft.
- Friction head loss along lateral, $H_f = 17.6$ ft.
- Elevation between pump and junction of lateral and main, $H_j = 0.75$ m
- Elevation between pump and water surface, $H_s = 4.2$ m
- Height of riser, $H_r = 25$ cm = 0.82 ft.
- Average operating pressure = 40 psi
- Lateral is laid on a level ground.

Other data are given as follows:

- ET = 8 mm/day
- Percolation = 3 mm/day
- Irrigation interval = 7 days
- Irrigation period = 12 hrs
- Pump efficiency = 80%
- Irrigation efficiency = 75%

Solution:

- a. The discharge, Q in m^3/sec , is

Solution:

$$Q = \frac{Ad_g}{t} = \frac{Ad_n}{t * \text{Eff}_a} = \frac{A(ET_{\text{crop}} + P)(II)}{t * \text{Eff}_a} = \frac{(10,000 \text{ m}^2) \left(\left(\frac{0.008 \text{ mm}}{\text{day}} \right) + \frac{0.003 \text{ mm}}{\text{day}} \right) (7 \text{ days})}{(12 \text{ hrs})(3600 \text{ sec})(0.75)} = 0.02377 \text{ m}^3/\text{s}$$

- b. The size of the pump is

Solution:

$$\text{BHP} = \frac{\gamma QTDH}{\text{Eff}_p} = \frac{\left(\frac{62.4\text{lb}}{\text{ft}^3}\right) \left(\frac{0.02377\text{m}^3}{\text{s}}\right) (128\text{ft})(3.28\text{ft})^3 (60\text{s})}{0.8 \left(\frac{33,000\text{ft} \cdot \text{lb}}{\text{min}}\right)} = 15.22\text{hp}$$

- c. The bulk density of a soil is 1.35 g/cc. What is the real specific gravity?

Solution:

$$A_s = 1.35$$

38. Given bed slope=0.0003; Manning's coefficient, n=0.03; b/d ratio=3.5; wetted perimeter, p=14; z=0.5. The velocity of flow along a trapezoidal canal is:

Solution:

$$\frac{b}{d} = 3.5; b = 3.5d$$

$$P = b + 2d\sqrt{1 + z^2}$$

$$14\text{m} = 3.5d + 2d\sqrt{1 + (0.5)^2}; 14 = 5.736d; d = 2.44\text{m} \text{ \& } b = (2.44)(3.5) = 8.54\text{m}$$

$$A = bd + zd^2 = (8.54\text{m})(2.44\text{m}) + (0.5)(2.44\text{m})^2 = 23.81\text{m}^2$$

$$R = \frac{A}{P} = \frac{23.81\text{m}^2}{14\text{m}} = 1.70$$

$$V = \frac{1}{n} R^{\frac{2}{3}} S^{\frac{1}{2}} = \frac{1}{0.03} (1.70\text{m})^{\frac{2}{3}} (0.0003)^{\frac{1}{2}} = \frac{0.82\text{m}}{\text{s}}$$

39. The thickness of a confined aquifer is 5 m, well diameter=40 cm, radius of influence=250 m, k=10 m/day, height of water in the pumping well=1,325 m, height of piezometric surface=1,500 m. The steady state discharge is:

Given:

Thickness = 5m

Well diameter = 40cm

Radius of influence = 250m

K = 10m/day

Height of water in well = 1,325m

Height of piezometric surface = 1,500m

Solution:

$$Q = \frac{2\pi k t (h_i - h_w)}{\ln\left(\frac{r_i}{r_w}\right)} = \frac{2\pi \left(\frac{10\text{m}}{\text{day}}\right) (5\text{m})(1500\text{m} - 1325\text{m})}{\ln\left(\frac{250\text{m}}{0.2\text{m}}\right) (86,400\text{sec})} = \frac{0.089\text{m}^3}{\text{s}}$$

40. How many sprinklers with spacing of 7m x 7m are needed to irrigate a rectangular field of 150m x 200m if the laterals are set parallel to the larger side of the field:

Solution:

No. of sprinklers per laterals = 200m/7m = 28.6 or 28 sprinklers

No. of lateral = 150m/7 = 21.43 or 21

No. of sprinklers = (28sprinklers)(21) = 588

41. If the theoretical flow of velocity in an orifice is 4 m/s, what is the height of water flowing above the center of the orifice?

Solution:

$$h = \frac{v^2}{2g} = \frac{\left(\frac{4\text{m}}{\text{s}}\right)^2}{2\left(\frac{9.81\text{m}}{\text{s}}\right)} = 0.81\text{m}$$

42. If the water strikes a point 0.5m horizontally and 100 mm vertically, what is the coefficient of discharge of the same orifice from problem #41.

Solution:

$$Q = (0.81\text{m}) (0.1) (4\text{m/s}) = 0.324\text{m}^3/\text{s}$$

$$Q = CA\sqrt{2gh}$$

$$C = \frac{Q}{A\sqrt{2gh}} = \frac{\left(\frac{0.324\text{m}^3}{\text{s}}\right)}{(0.1 \times 0.5)\sqrt{2(9.81)(0.81)}} = 1.62$$

43. Given an irrigation depth of 30 cm and 8hrs of irrigation time. Determine the discharge needed to irrigate a hectare of land.

Given: A = 1ha or 10,000m², depth = 0.3m, t = 8hrs

Solution:

$$Q = \frac{Ad_g}{t} = \frac{(10,000\text{m}^2)(0.3\text{m})}{8\text{hrs}} = \frac{375\text{m}^3}{\text{hr}}$$

44. A farmer is to irrigate a crop planted in furrows of 100m x 100m area. His furrow spacing is 50 cm and the application depth is 100 mm. If the inflow rate is 0.4 lps/100m, determine the time of irrigating each furrow:

Given:

$$\text{Area} = 10,000\text{m}^2$$

$$\text{Furrow spacing} = 50\text{cm}$$

$$\text{Application depth} = 100\text{mm}$$

$$\text{Inflow rate} = 0.4\text{lps}/100\text{m}$$

Solution:

$$t = \frac{Ad_g}{Q} = \frac{(10,000\text{m}^2)(0.1\text{m})\left(\frac{1,000\text{lit}}{\text{m}^3}\right)}{\left(\frac{0.4\text{lit}}{\text{s}}\right)\left(\frac{3600\text{s}}{1\text{hr}}\right)} = 694.44\text{hrs}$$

$$\text{No. of furrows} = \frac{100}{0.5} = 200 \text{ furrow}$$

$$\text{time for each furrow} = \frac{694.44\text{hrs}}{200\text{furrows}} = \frac{3.47\text{hr}}{\text{furrow}}$$

45. A rectangular lined canal will be used to irrigate 60 has. If a Francis weir of 0.4m crest will be used to control the flow, what would be the depth of water over the crest in order to supply 2lps/ha?

Given:

Note: Francis weir means rectangular weir

$$A = 60\text{has}$$

$$Q = 2\text{lps}/\text{ha}$$

Solution:

$$Q = 0.0184LH^{\frac{3}{2}} \text{ (for rectangular)}$$

$$H = \left(\frac{Q}{0.0184L}\right)^{\frac{2}{3}} = \left(\frac{2\text{lps}(60\text{m})}{0.0184(40\text{cm})}\right)^{\frac{2}{3}} = 30\text{cm or } 0.3\text{m}$$

46. If the canal in the problem #45 condition is changed into a triangular canal, what would be the depth of water over the crest if a V-notch weir is used?

Solution:

$$Q = 0.0138H^{\frac{5}{2}} \text{ (for triangular)}$$

$$H = \left(\frac{Q}{0.0138} \right)^{\frac{2}{5}} = \left(\frac{21 \text{ps}(60\text{m})}{0.0138} \right)^{\frac{2}{5}} = 37.65\text{cm or } 0.38\text{m}$$

47. The operating pressure of the sprinkler heads in a sprinkler irrigation system is 12.52 kg/cm². What is the allowable pressure (kg/m²) variation between the first and the last sprinkler head in a lateral?

Solution:

Note: Allowable pressure, head, or discharge variation in sprinkler or drip system is 20%.

$$\text{Allowable pressure variation} = (0.2) (12.52\text{kg/m}^2) = 2.504\text{kg/m}^2$$

48. If the speed of an irrigation centrifugal pump is increased by 30%, the power requirement is increased by

$$\text{Given: } N_2 = 1.3N_1$$

Solution:

$$\frac{P_2}{P_1} = \left(\frac{N_2}{N_1} \right)^3; P_2 = P_1 \left(\frac{N_2}{N_1} \right)^3 = P_1 \left(\frac{1.3N_1}{N_1} \right)^3 = P_1 \left(\frac{1.3N_1}{N_1} \right)^3 = 2.197P_1$$

49. A farmer collected a soil sample 2 days after irrigation. The sample has a diameter of 5 in., height of 8 in. and the sample weighed 3000 g before drying and 2340 after drying. Bulk density is g/cc.

Given:

$$D = 5\text{in}$$

$$H = 8\text{in}$$

$$\text{Initial weight} = 3000\text{g}$$

$$\text{Final weight} = 2340\text{g}$$

Solution:

$$V = \frac{\pi D^2 L}{4} = \frac{\pi (5\text{in})^2 (8\text{in})}{4} = 157.08\text{in}^3 \left(\frac{2.54\text{cm}}{1\text{in}} \right)^3 = 2,574.07\text{cm}^3$$

$$BD = \frac{2,340\text{g}}{2,574.07\text{cm}^3} = \frac{0.909\text{g}}{\text{cm}^3}$$

50. The field capacity in given condition in problem #49 is

Solution:

$$\text{Field capacity} = \frac{3000\text{g} - 2340\text{g}}{2340\text{g}} \left(\frac{12\text{in}}{\text{ft}} \right) = \frac{3.40\text{in}}{\text{ft}}$$

51. Assuming that the wilting point in the problem #49 is half of the field capacity, the readily available moisture in in/ft. is:

Solution:

$$\text{RAM} = (0.75) (3.4/2) = 1.269$$

52. A farm ditch has depth of flowing water of 60 cm, bottom width=50 cm, side slope=1 horizontal:1 vertical, channel slope=0.02%.

- a. The velocity of flow in m/sec is:

Note: for canals and ditches, $n = 0.02$

$$V = \frac{1}{n} R^{\frac{2}{3}} S^{\frac{1}{2}} = \frac{1}{0.02} (0.3\text{m})^{\frac{2}{3}} (0.0002)^{\frac{1}{2}} = \frac{0.317\text{m}}{\text{s}}$$

$$R = \frac{A}{P} = \frac{by + zy^2}{b + 2y\sqrt{1 + z^2}} = \frac{(0.5\text{m})(0.6\text{m}) + (1)(0.6\text{m})^2}{(0.5\text{m}) + 2(0.6\text{m})\sqrt{1 + (1)^2}} = 0.05\text{m}$$

- b. The discharge in m³/sec is

$$A = by + zy^2 = (0.5\text{m}) (0.6\text{m}) + (1) (0.6\text{m})^2 = 0.66\text{m}^2$$

- $Q = AV = (0.66\text{m}^2) (0.317\text{m/sec}) = 0.21\text{m}^3/\text{sec}$
53. Determine the size of a pump to irrigate a 5-ha rice-land. The water source is a stream with pumping head of 2 m. Evaporation=8mm/day, percolation=2mm/day, irrigation interval=5 days, irrigation period=10 hrs, friction loss=0, pump efficiency=80%, irrigation efficiency=80%.

Given:

Area = 5-ha
 Head = 2m
 Evaporation = 8mm/day
 Percolation = 2mm/day
 Irrigation interval = 5days
 Irrigation period = 10hrs
 Friction loss = 0
 Pump efficiency = 80%
 Irrigation efficiency = 80%

Solution:

$$\text{BHP} = \frac{\gamma QTDH}{\text{Eff}_p} = \frac{\left(\frac{1000\text{kg}}{\text{m}^3}\right) \left(\frac{0.0867\text{m}^3}{\text{s}}\right) (2\text{m})}{(0.8) \left(76.2\text{kg} \cdot \frac{\text{m}}{\text{s}}\right)} = 2.84\text{hp}$$

$$Q = \frac{Ad_g}{t} = \frac{A(ET_{\text{crop}} + P)(II)}{\text{Eff}_a * t} = \frac{(5 \times 10,000\text{m}^2) \left(\frac{0.008\text{mm}}{\text{day}} + \frac{0.002\text{mm}}{\text{day}}\right) (5\text{days})}{(0.80)(10\text{hrs})(3600\text{sec})} = \frac{0.0867\text{m}^3}{\text{s}}$$

54. A farmer proposes to use sprinkler on a rectangular field 160m x 120m with lateral line parallel to the shorter side. The land is adjacent to the river from which he pumps water. He needs a maximum application depth of 60 mm every 10 days. The maximum intake rate of the soil is 8.5 mm/hr and the application efficiency is 80%. Sprinkler spacing is 9m x 9m.

Given:

Area = 160mx120m
 Depth = 60mm every 10days
 Intake rate = 8.5mm/hr.
 Application efficiency = 80%
 Sprinkler spacing is 9mx9m

Solution:

- a. What is the desired discharge per sprinkler in lps?

Solution:

$$q = (9\text{m} \times 9\text{m}) \left(\frac{0.0085\text{m}}{\text{hr}}\right) \left(\frac{1,000\text{liters}}{\text{m}^3}\right) \left(\frac{1\text{hr}}{3,600\text{s}}\right) = 0.18125\text{lps}$$

- b. How many sprinklers are in a lateral?

Solution:

$$\text{No. of sprinklers} = 120\text{m}/9\text{m} = 13.33 \text{ or } 13 \text{ sprinklers}$$

- c. How many number of lateral positions?

$$\text{No. of laterals} = 160\text{m}/9\text{m} = 17.78 \text{ or } 17 \text{ laterals}$$

- d. How many hours will it take him to apply the water requirement in one lateral position?

Solution:

Note: if only one lateral, area is not included in the equation below.

$$t = \frac{d_g}{i * \text{Eff}_a} = \frac{(0.06\text{m})}{\left(\frac{0.0085\text{m}}{\text{hr}}\right) (0.8)} = 8.82\text{hrs}$$

- e. What is the required pump discharge in lps?

Solution:

$$Q_p = (160\text{m} \times 120\text{m}) \left(\frac{0.0085\text{m}}{\text{hr}} \right) \left(\frac{1\text{hr}}{3,600\text{s}} \right) \left(\frac{1,000\text{lit}}{\text{m}^3} \right) = 45.33\text{lps}$$

55. Water flows in a cylindrical tank at a rate of 20 m³/sec. If the radius of the tank is 2m, find how fast is the tank rising?

Give:

$$Q = 20\text{m}^3/\text{sec}$$

$$\text{Radius} = 2\text{m}$$

Solution:

$$Q = AV; V = \frac{Q}{A} = \frac{\frac{20\text{m}^3}{\text{s}}}{\pi(2\text{m})^2} = \frac{1.59\text{m}}{\text{s}}$$

56. Given the soil sampler with a diameter of 5.2 cm and length of 4.7 cm. What is the bulk volume of soil it could gather?

Solution:

$$V = \frac{\pi D^2 L}{4} = \frac{\pi(5.2\text{cm})^2(4.7\text{cm})}{4} = 99.8\text{cm}^3 \text{ or } 100\text{cm}^3$$

57. An engineer is to design the mainline of a sprinkler system which will use a steel (Scobey's coefficient $K = 0.40$) to carry a flow of 6.5lps. The length of pipe is 150m and the inside diameter is 100 mm. Compute the head loss of the system assuming a uniform flow.

Given:

$$\text{Scobey's coefficient } K = 0.40$$

$$Q = 6.5\text{lps}$$

$$\text{Length of pipe} = 150\text{m}$$

$$\text{Pipe diameter} = 100\text{mm}$$

Solution:

$$Q = \frac{K_s L Q^{1.9}}{D^{4.9}} (4.1 \times 10^6) = \frac{(0.40)(150\text{m})(6.5\text{lps})^{1.9}}{(100\text{mm})^{4.9}} (4.1 \times 10^6) = 1.37\text{m}$$

58. What is the evapotranspiration of corn during its growing period if the crop coefficient is 2.5 and the reference crop evapotranspiration is 6 mm/day?

Solution:

$$ET_a \text{ or } ET_{\text{crop}} = K_c ET_o = (2.5) \left(\frac{6\text{mm}}{\text{day}} \right) = \frac{15\text{mm}}{\text{day}}$$

59. The capacity of a certain canal will be measured using a Cipolletti weir having a length of 0.5m. If the height of the flowing water above the crest is 13 cm, what is the well capacity in lps?

Solution:

Note: Cipolletti weir is trapezoidal weir with discharge equation

$$Q = 0.0186 L H^{\frac{3}{2}} = 0.0186(50\text{cm})(13\text{cm})^{\frac{3}{2}} = 43.60\text{lps}$$

60. With this discharge, what will be the head above the crest if a V-notch is installed somewhere along the same canal?

Solution:

Note: V-notch discharge equation is

$$Q = 0.0138 H^{\frac{5}{2}}; H = \left(\frac{Q}{0.0138} \right)^{\frac{2}{5}} = \left(\frac{43.60\text{lps}}{0.0138} \right)^{\frac{2}{5}} = 25.1\text{cm}$$

61. Calculate the frictional head loss through mainline using following data:

Diameter of main pipe = 110mm

Rate of discharge flowing through mainline = 8lps

Length of pipe = 1000m

Hazen-William constant = 150 (for PVC)

Pipe thickness = 2.3mm

Solution:

$$H = \frac{\left(\frac{Q}{C}\right)^{1.852} * L}{D^{4.871}} = \frac{\left(\frac{8\text{lps}}{150}\right)^{1.852} * (1,000\text{m})}{(110\text{mm} - 2.3\text{mm} \times 2)^{4.871}} = 1.21 \times 10^{10} = 7.45\text{m}$$

62. Determine the discharge of individual sprinkler to apply the water at the rate of 10mm/hr. having spacing when the sprinkler spacing is 12m and lateral spacing is 12m.

Solution:

$$q = \frac{(12\text{m})(12\text{m}) \left(\frac{0.01\text{mm}}{\text{hr}}\right) (1,000\text{lit})}{3600\text{s}} = 0.4\text{lps}$$

63. Delivery of 360cfs to an 80-acre field is continued for 4 hrs. Tail water flow is estimated at 10cfs. Soil probing after the irrigation indicates that 1 ft. of water was stored in the root-zone. Compute the application efficiency.

Solution:

d_g = depth of water delivered to the farm

$$d_g = \frac{Q_t}{A} = \frac{\left(\frac{360\text{ft}^3}{\text{s}}\right) (4\text{hrs})(3600\text{s})}{80 \left(\frac{43560\text{ft}^2}{1\text{acre}}\right)} = 1.4876\text{ft}$$

$$\text{Eff}_a = \frac{1\text{ft}}{1.4876\text{ft}} = 67\%$$

64. How much water is needed in each irrigation if the moisture holding capacity of the soil is 1.5 in/ft. and irrigation was started when 40% is depleted? The crop uses 0.25 in/day of moisture and has a root depth of 3 ft. If there is no rain, how often will irrigation be required?

Given:

MHC = 1.5in/ft

AMD = 40%

CU = 0.25in/day

Root depth = 3ft

Solution:

$$d = (\text{AMD}) (\text{WHC}) (D) = (1.5\text{in/ft.})(0.4)(3\text{ft}) = 1.8\text{in}$$

$$t = d/\text{CU} = 1.8\text{in}/0.25\text{in/day} = 7.2 \text{ days}$$

65. Compute the water holding capacity of soil in inches if the F.C. is 30% and the wilting point is 15% of the dry soil density. The density is 80 lbs/ft³ and the soil depth to be irrigated is 2 feet.

Given:

FC = 30%

PWP = 15%

Density = 80lb/ft³

Soil depth = 2ft

Solution:

$$A_s = \frac{80}{62.4} = 1.282 \text{ Type equation here.}$$

$$\text{WHC} = (\text{FC} - \text{PWP}) A_s d_{rz} = (0.3 - 0.15)(1.282)(2\text{ft}) = 0.3846\text{ft or } 4.61\text{in}$$

66. The trapezoidal canal is to have a base width of 6 m and side slopes of 1:1. The velocity of flow is 0.6 m/sec. What slope must be given the bed in order to deliver 5 m³/sec. Use Kutther's C, n=0.025.

Solution:

$$Q = AV; A = Q/V = (5\text{m}^3/\text{sec}) / (0.6\text{m}/\text{sec}) = 8.33\text{m}^2$$

$$A = by + zy^2; 8.33\text{m}^2 = 6y + y^2; y = 1.163\text{m}$$

$$P = b + 2y\sqrt{1 + z^2} = 6 + 2(1.163)\sqrt{1 + 1^2} = 9.29\text{m}$$

$$R = A/P = 8.33\text{m}^2/9.29\text{m} = 0.897\text{m}$$

$$V = \frac{1}{n} R^{\frac{2}{3}} S^{\frac{1}{2}}; S = \left(\frac{nV}{R^{\frac{2}{3}}} \right)^2 = \left(\frac{(0.025) \left(\frac{0.6\text{m}}{\text{s}} \right)}{(0.897\text{m})^{\frac{2}{3}}} \right)^2 = 0.00026$$

67. What is the time required for water from the most remote region of the watershed to reach the outlet if the length of flow is 1550m and elevation difference between the outlet and the most remote region is 20m?

Given:

$$L = 1,550\text{m}$$

$$\text{Slope} = 20\text{m}$$

Solution:

Note: L should be in ft. and S is slope.

$$t = \frac{0.0078L^{0.77}}{S^{-0.385}} = \frac{0.0078L^{0.77}}{S^{-0.385}} = \frac{0.0078(1550 \times 3.28)^{0.77}}{\left(\frac{20\text{m}}{1550\text{m}} \right)^{-0.385}} = 29.73\text{min}$$

68. Given a probability of occurrence or exceedence of 5%. How many years will it take before a discharge of 1000 m³/sec will be equaled or exceeded?

Solution:

$$Y = 1/P = 1/0.05 = 20 \text{ years}$$

69. The field capacity and permanent wilting point for a given 0.8m root-zone soil are 35% and 10%. At a given time, the soil moisture in the given soil is 20% when a farmer irrigates the soil with 250mm depth of water. Assuming bulk specific gravity of the soil as 1.6, determine the amount of water wasted from the consideration of irrigation.

Solution:

$$D_n = (FC - PWP)A_s D_{rz} = (0.35 - 0.2)(1.6)(0.8\text{m}) = 0.192\text{m}$$

Therefore, the amount of water wasted is

$$= 0.250 - 0.192\text{m} = 0.58\text{mm or}$$

$$= 58/250 = 23.2\%$$

70. During a particular stage of the growth of a crop, consumptive use of water is 2.8mm/day. Determine the interval in days between irrigations, and depth of water to be applied when the amount of water available in the soil is 25% of the maximum depth of available water in the root zone which is 80mm. assume irrigation efficiency to be 65%.

$$CU = 2.8\text{mm}/\text{day}$$

$$\text{Irrigation efficiency} = 65\%$$

Solution:

$$II = \frac{d_n}{CU} = \frac{(80\text{mm})(1 - 0.25)}{\frac{2.8\text{mm}}{\text{day}}} = 21.43\text{days or } 21\text{days}$$

$$\text{Depth of water to be applied} = \frac{80\text{mm}(1-0.25)}{0.65} = 92.31\text{mm}$$

71. A well with a radius of 0.3m, including gravel envelope and developed zone, completely penetrates an unconfined aquifer with $K = 25\text{m/day}$ and initial water table at 30m above the bottom of the aquifer. The well is pumped so that the water level in the well remains at 22m above the bottom of the aquifer. Assuming that pumping has essentially no effect on the water table height at 300m from the well, determine the steady-state well discharge. Neglect well losses.

Solution:

$$Q = \frac{\pi k(h_i^2 - h_w^2)}{\ln\left(\frac{r_i}{r_w}\right)} = \frac{\pi\left(\frac{25\text{m}}{\text{day}}\right)(30\text{m}^2 - 22\text{m}^2)}{\ln\left(\frac{300\text{m}}{0.3\text{m}}\right)} = \frac{4,729.8\text{m}^3}{\text{day}}$$

Shape of cross-section	Area A	Wetted perimeter P	Hydraulic radius R	Water surface width T	Hydraulic depth D
Rectangle	$2h^2$	$4h$	$0.500h$	$2h$	h
Triangle (side slope 1 : 1)	h^2	$2.83h$	$0.354h$	$2h$	$0.500h$
Trapezoid (half of hexagon)	$1.73h^2$	$3.46h$	$0.500h$	$2.31h$	$0.750h$
Semicircle	$0.5 \pi h^2$	πh	$0.500h$	$2h$	$0.250 \pi h$
Parabola ($T = 2\sqrt{2h}$)	$1.89h^2$	$3.77h$	$0.500h$	$2.83h$	$0.667h$

72. Calculate the discharge rate of water of a suppressed rectangular weir 50 cm long with a head of 12 cm.

Solution:

$$Q = 0.0184 LH^{\frac{3}{2}} = 0.0184(50\text{cm})(12\text{cm})^{\frac{3}{2}} = 38.24\text{ps}$$

73. Calculate the discharge rate of water of a contracted rectangular weir 40 cm long with a head of 10 cm.

Solution:

$$Q = 0.0184 (L - 0.2H)H^{\frac{3}{2}} = 0.0184 (40\text{cm} - 0.2(10\text{cm}))(10\text{cm})^{\frac{3}{2}} = 22.1\text{ps}$$

74. Calculate the discharge rate of water of a 90° V-notch with a head of 12 cm.

Solution:

$$Q = 0.0138 H^{\frac{5}{2}} = 0.0138(12\text{cm})^{\frac{5}{2}} = 6.88\text{ps}$$

75. A persian wheel discharges at the rate of 11,000 litres per hour and works for eight hours each day. Estimate the area commanded by the water lift if the average depth of irrigation is 8 cm and irrigation interval is 15 days.

Given:

$Q = 11,000\text{lph}$ for 8hrs per day

Depth of irrigation = 8cm

Irrigation interval = 15days

Solution:

The total discharge for 15days:

$$V_1 = \frac{\left(\frac{11,000\text{lit}}{\text{hr}}\right)\left(\frac{8\text{hrs}}{\text{day}}\right)(15\text{days})}{1,000\text{lit}} = 1320\text{m}^3$$

The water required to cover one hectare to a depth of 8 cm

$$V_2 = 10,000\text{m}^3 * 0.08\text{m} = 800\text{m}^3$$

Area commanded by Persian wheel:

$$A = \frac{\text{Water available}}{\text{water required per unit area}} = \frac{1320\text{m}^3}{800\text{m}^3} = 1.65\text{ha}$$

76. A tube well with an average discharge of 10 liters per second irrigates one hectare cotton crop in 15 hours. Calculate the average depth of irrigation.

Solution:

$$V = \frac{\frac{10\text{liters}}{\text{s}} (15\text{hrs})(3600\text{s})}{1,000\text{liters}} = 540\text{m}^3$$

$$d = \frac{540\text{m}^3}{10,000\text{m}^2} = 0.054\text{m or } 5.4\text{cm}$$

77. Wheat crop requires 45 cm of irrigation water during crop season of 125 days. How much area can be irrigated with a flow of 18 liters per second for 10 hours each day?

Solution:

$$V = \frac{\left(\frac{18\text{liters}}{\text{s}}\right) \left(\frac{10\text{hrs}}{\text{day}}\right) (3600\text{s})(125\text{days})}{1,000\text{liters}} = 81,000\text{m}^3$$

$$\text{Area} = \frac{81,000\text{m}^3}{10,000\text{m}^2 * 0.45\text{m}} = 18\text{ha}$$

78. Find out net and gross irrigation requirement with the help of following data :

FC of soil = 21 %

Moisture content before irrigation = 7%

Depth of soil = 30 cm

Apparent density of soil = 1.5 Mg / m³

Field efficiency = 70 %

Solution:

$$d_n = (FC - PWP)A_s * d_{rz} = (0.21 - 0.07)(1.5)(0.3\text{m}) = 6.3\text{cm}$$

$$d_g = \frac{d_n}{E_a} = \frac{6.3\text{cm}}{0.7} = 9\text{cm}$$

79. After how many days will you apply water to soil if

FC of soil = 28 %

PWP = 13%

Apparent density of soil = 1.3 Mg / m³

Effective depth of root zone = 70 cm

Daily CU of water for the crop = 12 mm

Solution:

$$d_n = (FC - PWP)A_s * D_{rz} = (0.28 - 0.13)(1.3)(700\text{mm}) = 136.5\text{mm}$$

$$II = \frac{136.5\text{mm}}{12\text{mm}} = 11.375\text{days or } 11\text{days}$$

80. Calculate irrigation interval on the basis of following data

FC = 20 %

PWP = 8%

Bulk density = 1.4 Mg / m³

Root depth = 60 cm

ET rate = 0.5 cm / day

Allowable soil water depletion is equal to 25 per cent of available soil water.

Solution:

$$d_n = (FC - PWP)A_s * D_{rz} = (0.2 - 0.08)(1.4)(60\text{cm}) = 10.08\text{cm}$$

$$II = \frac{d_n}{ET} = \frac{10.08\text{cm} * 0.25}{\frac{0.5\text{cm}}{\text{day}}} = 5.04\text{days or 5days}$$

81. A stream of 135 litres/sec. was diverted from a canal and 100 litres /sec. were delivered to the field. An area of 1.6 ha was irrigated in 8 hours. The effective depth of root zone was 1.8 m. The run off loss in the field was 432 m³. The depth of water penetration varied linearly from 1.8 m at the head end of the field to 1.2 m at the tail end. Available moisture holding capacity of the soil is 20 cm/m depth of soil. Calculate water conveyance efficiency, water application efficiency, water storage efficiency and water distribution efficiency, irrigation was started at a moisture depletion level of 50 percent of the available moisture.

Given:

Stream = 135lps
 Q = 100lps
 Area = 1.6ha in 8hrs
 Root zone = 1.8m
 Runoff loss = 432m³
 Depth of water (linearly) = 1.8m
 Depth or water (end) = 1.2m
 AMWH = 20cm/m

Solutions:

- a. Conveyance efficiency

Solution:

$$Eff_c = \frac{\text{water from source}}{\text{water delivered}} = \frac{100\text{lps}}{135\text{lps}} = 74.07\%$$

- b. Application efficiency

Solution:

$$\text{Water delivered} = \frac{(100\text{lps})(8\text{hrs} \times 3600\text{s})}{1000\text{liters}} = 2,880\text{m}^3$$

$$\text{Water stored} = 2880 - 432 = 2448\text{m}^3$$

$$Eff_a = \frac{\text{water stored}}{\text{water delivered}} = \frac{2,448\text{m}^3}{2,880\text{m}^3} = 85\%$$

- c. Water storage efficiency

Solution:

$$\text{WHC of root-zone} = 20\text{cm/m} (1.8\text{m}) = 36\text{cm}$$

$$\text{Water need in root-zone} = 36\text{cm} - (36 \times 0.5) = 18\text{cm} = (0.18\text{m})(1.6\text{ha} \times 10,000\text{m}^2) = 2,880\text{m}^3$$

$$Eff_w = \frac{\text{water stored}}{\text{water needed}} = \frac{2,448\text{m}^3}{2,880\text{m}^3} = 85\%$$

- d. Water distribution efficiency

$$Eff_d = 100 \left(1 - \frac{y}{d} \right) = 100 \left(1 - \frac{0.3\text{m}}{1.5\text{m}} \right) = 80\%$$

$$d = \frac{1.8 + 1.2}{2} = 1.5\text{m}$$

$$y = \frac{(1.8\text{m} - 1.5\text{m}) + (1.5\text{m} - 1.2\text{m})}{2} = 0.3\text{m}$$

82. A volume of undisturbed soil, 0.015 m³, obtained from a construction site has wet mass of 31 kg and a dry mass of 27.5 kg. Gs of the soil solids is 2.71. Determine the void ratio, water content, and the degree of saturation for the in place condition.

Solution:

$$BD = \frac{27.5\text{kg} \times 1000\text{g}}{0.015\text{m}^3 (100\text{cm})^3} = \frac{1.83\text{g}}{\text{cm}^3}$$

$$n = 1 - \frac{BD}{PD} = 1 - \frac{1.83}{2.71} = 0.3247 \text{ or } 32.47$$

$$\text{void ratio } (e) = \frac{n}{1 - n} = \frac{0.32}{1 - 0.32} = 0.48$$

$$W_c = \frac{FW - OD}{OD} = \frac{31\text{kg} - 27.5\text{kg}}{27.5\text{kg}} = 12.72\%$$

$$\text{degree of saturation} = \frac{W_c * SG}{e} = \frac{(0.1272)(2.71)}{0.48} = 72\%$$

83. A sample of clay soil taken from a construction site has a wet mass of 45.6 kg and a related in place volume of 0.021 m³. After being dried in an oven, the dry soil mass is 39.1 kg. The specific gravity of the soil particles is 2.73. Determine the soil moisture content, void ratio, and the degree of saturation.

Solution:

$$MC = \frac{FC - OD}{OD} = \frac{45.6\text{kg} - 39.1\text{kg}}{39.1\text{kg}} = 16.62\%$$

$$BD = \frac{39.1\text{kg} \times 1000\text{g}}{0.021(100\text{cm})^3} = \frac{1.86\text{g}}{\text{cm}^3}$$

$$n = 1 - \frac{BD}{PD} = 1 - \frac{1.86}{2.73} = 31.87\%$$

$$\text{void ratio } (e) = \frac{n}{1 - n} = \frac{0.3187}{1 - 0.3187} = 0.47$$

$$\text{degree of saturation} = \frac{W_c * SG}{e} = \frac{(0.1662)(2.73)}{0.47} = 97\%$$

84. A soil sample is taken from a small test hole in a soil borrow pit. The volume of the test hole is determined to be 0.028 m³. The wet mass of the soil is 56 kg and the dry mass is 49 kg. The specific gravity of the soil solids is 2.72. Determine the wet and dry density, wet and dry unit weight, moisture content, void ratio and degree of saturation. Ans 2 Mg/m³ or 124.8 pcf, 1.75 Mg/m³ or 109.2 pcf, 14.3%, 0.556, 70%

Solution:

$$BD = \frac{(49\text{kg})(1000\text{g})}{(0.028\text{m}^3)(100\text{cm})^3} = \frac{1.75\text{g}}{\text{cm}^3}$$

$$DD = \frac{(56\text{kg})(1000\text{g})}{(0.028\text{m}^3)(100\text{cm})^3} = \frac{2\text{g}}{\text{cm}^3}$$

$$W_w = \frac{56 - 49}{56} = 12.5\%$$

$$W_d = \frac{56 - 49}{49} = 14.3\%$$

$$\text{Porosity } (n) = 1 - \frac{1.75}{2.72} = 0.357$$

$$\text{void ratio } (e) = \frac{n}{1 - n} = \frac{0.357}{1 - 0.357} = 0.555 \text{ or } 55.5\%$$

$$\text{degree of saturation} = \frac{W_c * SG}{e} = \frac{(0.143)(2.72)}{0.555} = 70\%$$

85. An undisturbed soil sample has a dry mass of 59 kg and an in situ (in ground) volume of 0.035 m³. The specific gravity of the soil particles is 2.65. Determine the void ratio

Solution:

$$BD = \frac{(59\text{kg})(1000)}{(0.035\text{m}^3)(100\text{cm})^3} = \frac{1.686\text{g}}{\text{cm}^3}$$

$$\text{Porosity } (n) = 1 - \frac{BD}{PD} = 1 - \frac{1.686}{2.65} = 36.38\%$$

$$\text{void ratio } (e) = \frac{n}{1 - n} = \frac{0.3638}{1 - 0.3638} = 57.20\%$$

86. A sharp edged cylinder 150 mm in diameter is carefully driven into the soil so that negligible compaction occurs. A 200 mm column of soil is secured. The wet weight is 5525 g and the dry weight is 4950 g. What is the moisture content of the soil sample in a dry and wet weight basis? The apparent specific gravity and the moisture content in volumetric basis? Ans 11.62%, 10.41%, 1.4, 16.27%

Solution:

$$MC_w = \frac{5525\text{g} - 4950\text{g}}{5525\text{g}} = 10.41\%$$

$$MC_d = \frac{5525\text{g} - 4950\text{g}}{4950\text{g}} = 11.62\%$$

$$BD = \frac{4950\text{g}}{\pi \left(\frac{150}{10}\right)^2 \left(\frac{200}{10}\right)} = \frac{1.40\text{g}}{\text{cm}^3}$$

$$A_{SG} = \frac{BD}{WD} = \frac{1.40}{1} = 1.40$$

$$\text{porosity } (n) = 1 - \frac{BD}{PD} = 47.20\%$$

$$\text{void ratio } (e) = \frac{0.472}{1 - 0.472} = 0.894$$

$$MC_{\text{volume}} (\text{degree of saturation}) = \frac{0.1021 * 1.40}{0.894} = 16.30$$

87. The soil moisture content at field capacity is 27% (dry weight basis) and the moisture content at the time of irrigation is 19.0%. The apparent specific gravity is 1.3 and the depth of soil to be wetted is 1.0 m. how many ha-mm per ha of water must be applied? How long will it take to irrigate the 5 hectares with 120lps stream?

Given:

$$FC = 27\%$$

$$MC = 19\%$$

$$ASG = 1.3$$

$$\text{Depth of soil} = 1\text{m}$$

$$\text{Area} = 5\text{ha}$$

$$Q = 120\text{lps}$$

Solution:

$$IA = \frac{(0.27 - 0.19)(1000\text{mm})(1.3)(1\text{ha})}{1\text{ha}} = \frac{104\text{ha} - \text{mm}}{\text{ha}}$$

$$t = \frac{Ad}{Q} = \frac{(5 \times 10,000\text{m}^2)(0.104\text{mm})(1000\text{lit})}{120\text{lps}(3600)} = 12.04\text{hr}$$

88. A farmer desires to irrigate a border which is 40 m wide and 150 m long. He wants to apply a stream of 60 lps into his field in 1 hour. How much amount of water did he apply to his field in ha-mm/ha or mm?

Given:

$$\text{Area} = 40\text{m} \times 150\text{m}$$

$$Q = 60\text{lps}$$

$$\text{Time} = 1\text{hr}$$

Solution:

$$d = \frac{Qt}{A} = \frac{(60\text{lps})(1\text{hr})(3600\text{s})(1\text{m}^3)(1000\text{mm})}{(40\text{m})(150\text{m})(1000\text{lit})} = 36\text{mm}$$

89. The bottom width of a trapezoidal canal is 4 m and the side slopes are 2H to 1V. Find the hydraulic radius, in meters, when the water flows 1.25 m deep.

Solution:

$$R = \frac{by + zy^2}{b + 2y\sqrt{1 + z^2}} = \frac{(4\text{m})(1.25\text{m}) + (2)(1.25)^2}{4\text{m} + 2(1.25\text{m})\sqrt{1 + 2^2}} = 0.847\text{m}$$

90. The discharge through a Cipolletti weir is 0.52 m³/s. the head on it is one fourth of the crest's length. What is the head in m?

Solution:

$$\text{Since, } H = \frac{1}{4} L$$

$$Q = 0.0186 L H^{\frac{3}{2}} = 0.0186 L \left(\frac{L}{4}\right)^{\frac{3}{2}} = 0.002325 L^{\frac{5}{2}}$$

$$L = \left(\frac{Q}{0.002325}\right)^{\frac{2}{5}} = \left(\frac{\frac{520\text{Lit}}{\text{s}}}{0.002325}\right)^{\frac{2}{5}} = 137.985\text{cm}$$

$$H = \frac{137.985\text{cm}}{4} = 34.49\text{cm or } 0.34\text{m}$$

91. For a triangular notch weir, what is the discharge when the depth of water vertically above the apex of the weir notch is 20cm at a point 1.5 m upstream from the weir?

Solution:

Note: standard coefficient of discharge of triangular v-notch is $C = 0.499$

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} H^{\frac{5}{2}} = \frac{8}{15} (0.469) \sqrt{2(9.81)} \tan \frac{90}{2} (0.2\text{m})^{\frac{5}{2}} = \frac{0.020\text{m}^3}{\text{s}} \text{ or } 20\text{lps}$$

92. If the head over an 80° triangular V-notch weir is 300 mm, determine the discharge. Assume $C = 0.6$. Ans 58.63 L/s

Solution:

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} H^{\frac{5}{2}} = \frac{8}{15} (0.6) \sqrt{2(9.81)} \tan \frac{80}{2} (0.3\text{m})^{\frac{5}{2}} = \frac{0.05863\text{m}^3}{\text{s}} \text{ or } 58.63\text{lps}$$

93. Determine the head on a suppressed rectangular weir if $Q = 4 \text{ m}^3/\text{s}$ $L = 3\text{m}$ and $C_w = 0.62$

Solution:

$$Q = \frac{2}{3} C_d \sqrt{2g} L H^{\frac{3}{2}} \text{ For } C_d; Q = C_w L H^{\frac{3}{2}} \text{ of } C_w$$

$$Q = C_w L H^{\frac{3}{2}}$$

$$H = \left(\frac{Q}{C_w L} \right)^{\frac{2}{3}} = \left(\frac{\frac{4\text{m}^3}{\text{s}}}{(0.62)(3\text{m})} \right)^{\frac{2}{3}} = 1.67\text{m}$$

94. A trapezoidal canal having a base width of 1.8 m and a top of 2.8 m is 1.2 m deep. If the normal flow is 17.857 m³/s at 1.2 m deep and a coefficient of roughness $n = 0.016$, compute the hydraulic slope

Given:

Base = 1.8m

Top width = 2.8m

$y = 1.2\text{m}$

$n = 0.016$

$Q = 17.857\text{m}^3/\text{s}$

Solution:

$$z = 1.2/2.8 = 0.43$$

$$A = by + zy^2 = (1.8\text{m})(1.2\text{m}) + (0.43)(1.2\text{m})^2 = 2.78\text{m}^2$$

$$P = b + 2y\sqrt{1 + z^2} = 1.8\text{m} + 2(1.2\text{m})\sqrt{1 + (0.43)^2} = 4.41\text{m}$$

$$R = \frac{A}{P} = \frac{2.78\text{m}^2}{4.41\text{m}} = 0.63\text{m}$$

$$Q = \frac{1}{n} R^{\frac{2}{3}} S^{\frac{1}{2}} A$$

$$S = \left(\frac{Qn}{AR^{\frac{2}{3}}} \right)^2 = \left(\frac{\left(\frac{17.857\text{m}^3}{\text{s}} \right) (0.016)}{(2.78\text{m}^2)(0.63\text{m})^{\frac{2}{3}}} \right)^2 = 0.02$$

95. A trapezoidal canal has a bottom width of 2.5 m and its sides are inclined 60° with horizontal. It can carry water up to a depth of 3 m. $S = 0.0008$ and $n = 0.03$. Find the discharge in m³/s when flowing full. Use the Manning's equation.

Given:

Base = 2.5m

$$z = \frac{1}{\sqrt{3}} = 0.577$$

$y = 3\text{m}$

$S = 0.0008$

$n = 0.03$

Solution:

$$Q = \frac{1}{n} R^{\frac{2}{3}} S^{\frac{1}{2}} A$$

$$A = by + zy^2 = (2.5\text{m})(3\text{m}) + (0.577)(3\text{m})^2 = 12.693\text{m}^2$$

$$P = b + 2y\sqrt{1 + z^2} = 2.5\text{m} + 2(3\text{m})\sqrt{1 + (0.577)^2} = 9.43\text{m}$$

$$R = 12.693\text{m}^2 / 9.43\text{m} = 1.346\text{m}$$

$$Q = \frac{1}{n} R^{\frac{2}{3}} S^{\frac{1}{2}} A = \frac{1}{0.03} (1.346\text{m})^{\frac{2}{3}} (0.0008)^{\frac{1}{2}} (12.693\text{m}^2) = \frac{14.60\text{m}^3}{\text{s}}$$

96. the bottom width of the most hydraulically efficient concrete-lined rectangular channel with a bottom slope of 0.1% to carry a discharge of 1.5m³/s is

Solution:

Note: $n = 0.015$ standard for designing canal.

$$AR^{\frac{2}{3}} = \frac{Qn}{S^{\frac{1}{2}}} = \frac{\left(\frac{1.5\text{m}^3}{\text{s}}\right)(0.015)}{(0.001)^{\frac{1}{2}}} = 0.70$$

$$(2y^2)\left(\frac{y}{2}\right)^{\frac{2}{3}} = 0.70$$

$$\frac{2y^{\frac{8}{3}}}{1.587} = 0.70$$

$$y = \left(\frac{0.70 \times 1.587}{2}\right)^{\frac{3}{8}} = 0.8$$

$$b = 2y = 2(0.8) = 1.6\text{m}$$

97. Water is flowing in a 15 cm ID pipe at a velocity of 0.3 m/s. The pipe enlarges to an inside diameter of 30 cm. What is the velocity in the larger section, the volumetric flow rate, and the mass flow rate?

Given:

$$\text{ID1} = 15\text{cm}$$

$$V1 = 0.3\text{m/s}$$

$$\text{ID2} = 30\text{cm}$$

$$V2 = ?$$

Solution:

$$A_1 V_1 = A_2 V_2; V_2 = \frac{A_1 V_1}{V_1} = \frac{\left(\frac{\pi(0.15\text{m})^2}{4}\right)\left(\frac{0.3\text{m}}{\text{s}}\right)}{\frac{\pi(0.3\text{m})^2}{4}} = 0.075\text{m/s}$$

Volumetric flow rate:

$$Q = \frac{\pi(0.15\text{m})^2\left(\frac{0.3\text{m}}{\text{s}}\right)}{4} = \frac{0.0053\text{m}^3}{\text{s}}$$

Mass flow rate:

$$\dot{m} = Q\rho_w = \left(\frac{0.0053\text{m}^3}{\text{s}}\right)\left(\frac{1,000\text{kg}}{\text{m}^3}\right) = \frac{5.3\text{kg}}{\text{s}}$$

98. A rectangular channel 6.10 m wide has a depth of 90 cm of water flowing with a mean velocity of 0.75 m/s. Determine the height of standard sharp-crested weir if the depth in the channel of approach is 1.50 m.

Given:

$$\text{Width} = 6.10\text{m}$$

$$\text{Depth} = 90\text{cm}$$

$$\text{Velocity} = 0.75\text{m/s}$$

Solution:

$$Q = (6.10\text{m})(0.9\text{m})\left(\frac{0.75\text{m}}{\text{s}}\right) = \frac{6.8625\text{m}^3}{\text{s}}$$

$$H = \left(\frac{\frac{4.1175\text{m}^3}{\text{s}}}{1.84(6.10\text{m})}\right)^{\frac{2}{3}} = 0.51$$

99. A pipe drainage system is to be designed for the following conditions:

Drainage coefficient ----- 8mm/day

Drain installation depth ----- 1.5m
 Maximum depth to water table ----- 1m (bellow the ground surface)
 Depth of impervious layer from the ground surface – 4m
 Average hydraulic conductivity ----- 1.5m/day

Assuming steady state conditions, the drain spacing should be

Solution:

$$S^2 = \frac{4KL(H + 2D)}{R}$$

$$S = \sqrt{\frac{4\left(\frac{1.5\text{m}}{\text{day}}\right)(0.5\text{m})(2(3\text{m}) - 0.5\text{m})}{\frac{0.008\text{m}}{\text{day}}}} = 45\text{m}$$

100. A floodway has been designed for discharge with return period of 75 years. The risk that the structure will experience the design flood one or more times in 20 years.

Solution:

$$P = \left(1 - \left(1 - \frac{1}{T}\right)^n\right) = \left(1 - \left(1 - \frac{1}{75}\right)^{20}\right) = 23.5\%$$

101. If 16 is 4 more than 4x, find 5x-1

Solution:

$$16 = 4x + 4$$

$$x = 3$$

$$5x - 1 = 5(3) - 1 = 14$$

102. Find the value of x in $\frac{x+1}{3} + \frac{2x}{4} = 47 - 2x$.

Solution:

$$\left(\frac{x+1}{3} + \frac{2x}{4} = 47 - 2x\right)(12)$$

$$4x + 4 + 6x = 564 - 24x$$

$$x = 16.47$$

103. Find the value of w in the following equations:

$$3x - 2y + w = 11$$

$$x + 5y - 2w = -9$$

$$2x + y - 3w = -6$$

Solution:

Multiply 2nd equation by 3.

$$3x + 15y - 6w = -27 \text{ Equation (4)}$$

Subtract (4) by (1) and you will get

$$y = \frac{7y - 38}{17} \text{ (Equation 5)}$$

Multiply (2) by 2 and the result is:

$$2x + 10y - 4w = -18 \text{ (Equation 6)}$$

Subtract equation (6) by (3) and you will get:

$$9y - w = -12 \text{ (Equation 7)}$$

Substitute equation (5) in (7) and you get the value of w:

$$w = 3$$

104. simplify $\frac{(x^2y^3z^{-2})^{-3}(x^{-3}yz^3)^{-\frac{1}{2}}}{(xyz^{-3})^{-\frac{5}{2}}}$

Solution:

$$\frac{(x^2y^3z^{-2})^{-3}(x^{-3}yz^3)^{-\frac{1}{2}}}{(xyz^{-3})^{-\frac{5}{2}}} = \frac{(x^{-6}y^{-9}z^6)\left(x^{\frac{3}{2}}y^{-\frac{1}{2}}z^{-\frac{3}{2}}\right)}{x^{-\frac{5}{2}}y^{-\frac{5}{2}}z^{\frac{15}{2}}} = x^{-6+\frac{3}{2}+\frac{5}{2}}y^{-9-\frac{1}{2}+\frac{5}{2}}z^{6-\frac{3}{2}-\frac{15}{2}}$$

$$= x^{-2}y^{-7}z^{-3} \text{ or } \frac{1}{x^2y^7z^3}$$

105. simplify the following equation

$$\frac{5x}{2x^2+7x+3} - \frac{x+3}{2x^2-3x-2} + \frac{2x+1}{x^2+x-6}$$

Solution:

$$\frac{5x}{(2x+1)(x+3)} - \frac{x+3}{(2x+1)(x+3)} + \frac{2x+1}{(x+3)(x-2)}$$

$$= \frac{5x(x-2) - (x+3)(x+3) + (2x+1)(2x+1)}{(2x+1)(x+3)(x-2)} = \frac{4}{x+3}$$

106. simplify $\left\{x^{\frac{2}{3}}\left\{x^{-\frac{1}{3}}y^{-\frac{1}{2}}(x^2y^{-2})^{-\frac{2}{3}}\right\}^{\frac{1}{2}}\right\}^6$

Solution:

$$x^4\left[x^{-\frac{1}{3}}y^{-\frac{1}{2}}x^{-\frac{4}{3}}y^{-\frac{4}{3}}\right]^3 = x^4\left[x^{-\frac{5}{3}}y^{\frac{5}{6}}\right]^3 = x^4\left[x^{-5}y^{\frac{5}{2}}\right] = x^{-1}y^{\frac{5}{2}} = \frac{y^{\frac{5}{2}}}{x}$$

107. simplify $7^{a+2}-8(7)^{a+1}+5(7)^a+49(7)^{a-2}$

Solution:

$$7^{a+2}-8(7)^{a+1}+5(7)^a+49(7)^{a-2} = 7^a7^2-8(7)^a7^1+(5)7^a + \frac{49(7)^a}{7^2}$$

$$7^a(49-56+5+1) = -7^a$$

108. solve for y: $\frac{x}{(b-c)} = \frac{y}{(a-c)} = \frac{z}{(a-b)}$

Solution:

$$\frac{x}{(b-c)} = \frac{y}{(a-c)} = \frac{z}{(a-b)}$$

$$x = \frac{y(b-c)}{a-c} = \frac{by-cy}{a-c}$$

$$z = \frac{y(a-b)}{a-c} = \frac{ay-by}{a-c}$$

$$x+z = \frac{by-cy}{a-c} + \frac{ay-by}{a-c} = \frac{ay-cy}{a-c}$$

$$x+z = y$$

109. Find the value of A in the equation: $\frac{x^2+4x+10}{x^3+2x^2+5x} = \frac{A}{x} + \frac{B(2x+2)}{x^2+2x+5} + \frac{C}{x^2+2x+5}$

Solution:

$$\frac{x^2+4x+10}{x^3+2x^2+5x} = \frac{A}{x} + \frac{B(2x+2)}{x^2+2x+5} + \frac{C}{x^2+2x+5} = \frac{A(x^2+2x+5) + Bx(2x+2) + Cx}{x(x^2+2x+5)}$$

$$\frac{x^2+4x+10}{x^3+2x^2+5x} = \frac{Ax^2+2Ax+5A+2Bx^2+2Bx+Cx}{x^3+2x^2+5x}$$

by equating constants: $10 = 5A$; $A = 2$

110. Evaluate: $64^x \cdot 4^y$

Solution:

$$64^x \cdot 4^y = (4^3)^x (4)^y = 4^{3x} \cdot 4^y = 4^{3x+y}$$

111. Solve for x in the following equations:

$$27^x = 9^y$$

$$81^y 3^{-x} = 243$$

Solution:

$$81^y 3^{-x} = 243$$

$$(3)^{4y} 3^{-x} = (3)^5 \text{ (Equation 1)}$$

$$27^x = 9^y$$

$$(3)^{3x} = (3)^{2y}$$

squaring both sides

$$(3)^{6x} = (3)^{4y} \text{ (Equation 2)}$$

Substitute (2) in (1):

$$(3)^{6x} (3)^{-x} = (3)^5; 6x - x = 5; \mathbf{x = 1}$$

112. If $p - q = 5$ and $pq = \frac{k}{2}$, then $p^2 + q^2$ equals

Solution:

$$(p - q)^2 = 5^2$$

$$p^2 - 2pq + q^2 = 25$$

$$p^2 + q^2 = 25 + 2pq = 25 + 2\left(\frac{k}{2}\right) = \mathbf{25 + k}$$

113. find the value of x which will satisfy the following expression: $\sqrt{x - 2} = \sqrt{x} + 2$

Solution:

Squaring both sides:

$$x - 2 = x + 4\sqrt{x} + 4$$

$$4\sqrt{x} = -6$$

$$\sqrt{x} = -\frac{3}{2} \text{ (Squaring both sides again)}$$

$$\mathbf{x = \frac{9}{4}}$$

114. simplify $\sqrt{\frac{ab}{\sqrt[3]{ab}}}$

Solution:

$$\sqrt{\frac{ab}{\sqrt[3]{ab}}} = \sqrt{\frac{ab}{(ab)^{\frac{1}{3}}}} = \sqrt{(ab)^{\frac{2}{3}}} = \left((ab)^{\frac{2}{3}}\right)^{\frac{1}{2}} = (ab)^{\frac{1}{3}} = \mathbf{\sqrt[3]{ab}}$$

115. solve for x: $\sqrt{x + 2\sqrt{2x + 3}} - 3 = 0$

Solution:

$$\sqrt{x + 2\sqrt{2x + 3}} = 3 \text{ (Squaring both sides)}$$

$$x + 2\sqrt{2x + 3} = 9$$

$$2\sqrt{2x + 3} = 9 - x \text{ (squaring both sides)}$$

$$8x + 12 = 81 - 18x + x^2; x^2 - 26x = -69$$

$$(x + 13)^2 = -69 + (13)^2 = 100$$

$$(x + 13)^2 = 100; \mathbf{x = 3}$$

116. The polynomial $x^3 + 4x^2 - 3x + 8$ is divided by $x - 5$, then the remainder is

Solution:

Note: using the remainder theorem, remainder = $f(5)$

$$\text{Remainder} = (5)^3 + 4(5)^2 - 3(5) + 8 = 218$$

117. The mean proportional of 4 and 36.

Solution:

$$\frac{4}{x} = \frac{x}{36}; x^2 = 144 = \sqrt{144} = 12$$

118. The arithmetic mean of 6 numbers is 17. If two numbers are added to the progression, the new set of numbers will have an arithmetic mean of 19. What are the two numbers if their difference is 4?

Solution:

Let x : the first number

$x + 4$ = the second number

y = some of the original 6 numbers

$$\text{Arithmetic mean} = \frac{y}{6} = 17; y = 17 * 6 = 102$$

$$\frac{y + x + (x + 4)}{6 + 2} = 19$$

$$\frac{102 + 2x + 4}{8} = 19$$

$$x = 23$$

$$x + 4 = 27$$

119. The arithmetic mean of 80 numbers is 55. If two numbers namely 250 and 850 are removed, what is the arithmetic mean of the remaining numbers?

Solution:

Let x = the arithmetic sum of 80 numbers,

$$\text{Arithmetic mean} = \frac{x}{80} = 55; x = 80 * 55 = 4400$$

y = new arithmetic mean

$$= \frac{x - (250 + 850)}{80 - 2} = 42.31$$

120. If 20 bakers can bake 40 pizzas in 8hrs, how many bakers can bake 10 pizzas in 2hrs?

Solution:

$$\text{Rate} = \frac{20 \text{ bakers}(8\text{hrs})}{40 \text{ pizzas}} = \left(\frac{4 \text{ baker} - \text{hr}}{\text{pizza}} \right) \left(\frac{10 \text{ pizzas}}{2 \text{ hrs}} \right) = 20 \text{ bakers}$$

121. Mary is 24 years old. Mary is twice as old as Ann was when Mary was as old as Ann is now. How old is Ann now?

Solution:

	Past	Present
Marry	X	24
Ann	12	X

$$24 - x = x - 12; x = 18$$

122. The sum of Kim's and Kevin's ages is 18. In 3 years, Kim will be twice as old as Kevin. What are their ages now?

Solution:

	Past	Present
Kevin	X	X+3

Kim	Y	Y+3
-----	---	-----

$$x + y = 18; y = 18 - x \text{ (Equation 1)}$$

$$y = 2x$$

$$y + 3 = 2(x + 3) \text{ (Equation 2)}$$

Substitute 1 to 2:

$$x = 5 \text{ years old}$$

$$y = 13 \text{ years old}$$

123. Robert is 15 years older than his brother Stan. However “y” years ago, Robert was twice as old as Stan. If Stan is now “b” years old and $b > y$, find the value of $(b - y)$.

Solution:

	Past	Present
Robert	$b + 15 - y$	$b + 15$
Stan	$b - y$	b

$$b + 15 - y = 2(b - y); b - y = 15$$

124. JJ is three times as old as Jan-Jan. Three years ago, JJ was four times as old as Jan-Jan. The sum of their ages is

Solution:

	Past	Present
JJ	$3x - 3$	$3x$
Jan-Jan	$x - 3$	x

$$3x - 3 = 4(x - 3); x = 9 \text{ yrs old}; 3x = 3(9) = 27 \text{ years old}; \text{sum of their ages} = 36 \text{ yrs.}$$

125. A girl is one-third as old as her brother and 8 years younger than her sister. The sum of their ages is 38 years. How old is the girl?

Solution:

Let x be the age of the girl

y be the age of her brother

z be the age of her sister

$$y = 3x; z = x + 8; x + y + z = 38$$

$$x + (3x) + (x + 8) = 38; x = 6 \text{ years old}$$

126. Paula is now 18 years old and his colleague Monica is 14 years old. How many years ago was Paula twice as old as Monica?

Solution:

	Past	Present
Paula	$18 - x$	18
Monica	$14 - x$	14

$$18 - x = 2(14 - x); x = 10 \text{ years ago}$$

127. A father tells his son “I was your age now when you were born”. If the father is now 38 years old, how old was his son 2 years ago?

Solution:

	Past	Present
Father	x	38
Son	0	x

$$38 - x = x - 0; x = 19 \text{ yrs old}; \text{two years ago, the son was } (19 - 2) = 17 \text{ years old}$$

128. Six years ago, Nilda was five times as old as Riza. In five years, Nilda will be three times as old as Riza. What is the present age of Riza?

Solution:

	Past	Present	Future
Nilda	$5(x - 6)$	?	$3(x + 5)$
Riza	$X - 6$	X	$X + 5$

$$3(x + 5) - 5(x - 6) = x + 5 - (x - 6); x = 17 \text{ yeras old}$$

129. A pump can pump out water from a tank in 11 hours. Another pump can pump out water from the same tank in 20 hours. How long will it take both pumps to pump out the water in the tank?

Solution:

Let x = the time needed to complete the work.

$$\frac{1}{20} + \frac{1}{11} = \frac{1}{x}; x = 7.096 \text{ hours}$$

130. A 400mm diameter pipe can fill the tank alone in 5 hours and another 600mm diameter pipe can fill the tank alone in 4 hours. A drain pipe 300mm diameter can empty the tank in 20 hours. Will all the three pipes open, how long will it take the tank?

Solution:

$$\frac{1}{5} + \frac{1}{4} - \frac{1}{20} = \frac{1}{x}; x = 2.5 \text{ hours}$$

131. A tank is filled with an intake pipe in 2 hours and emptied by an outlet pipe in 4 hours. If both pipes are opened, how long will it take to fill the empty tank?

Solution:

$$\frac{1}{2} - \frac{1}{4} = \frac{1}{x}; x = 4 \text{ hrs}$$

132. A tank can be filled in 9 hours by one pipe. 12 hours by a second pipe and can be drained when full by a third pipe in 15 hours. How long will it take to fill an empty tank with all the pipes in operation?

Solution:

$$\frac{1}{9} + \frac{1}{12} - \frac{1}{15} = \frac{1}{x}; x = 7.826 \text{ hrs or } 7 \text{ hours and } 50 \text{ minutes}$$

133. Pedro can paint a fence 50% faster than Juan and 20% faster than Pilar, and together they can paint a given fence in 4 hours. How long will it take Pedro to paint the same fence if he had to work alone?

Solution:

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = \frac{1}{4} \text{ (Equation 1)}$$

$$\frac{1}{A} = 1.5 \left(\frac{1}{B} \right); \frac{1}{B} = 0.666 \left(\frac{1}{A} \right) \text{ (Equation 2)}$$

$$\frac{1}{A} = 1.2 \left(\frac{1}{C} \right); \frac{1}{C} = 0.833 \left(\frac{1}{A} \right) \text{ (Equation 3)}$$

Substitute (2) and (3) in (1):

$$\frac{1}{A} + 0.666 \left(\frac{1}{A} \right) + 0.833 \left(\frac{1}{A} \right) = \frac{1}{4}; A = 10 \text{ hours}$$

134. Glenn can paint a house in 9 hours while Stewart can paint the same house in 16 hours. They can work together for 4 hours. After 4 hours, Stewart left and Glenn finished the job alone. How many more days did it take Glenn to finish the job?

Solution:

Note: (rate) (time) = 1 (complete job)

$$4 \left(\frac{1}{9} + \frac{1}{16} \right) + \frac{1}{9} (x) = 1; x = 2.75 \text{ hours}$$

135. It takes Butch twice as long as it takes Dan to do a certain piece of work. Working together they can do the work in 6 days. How long would it take Dan to do it alone?

Solution:

Let x = time for Butch to finish a certain job working alone

y = time for Dan to finish a certain job working alone

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{6} \text{ (Equation 1)}$$

$$x = 2y \text{ (Equation 2)}$$

Substitute (2) in (1):

$$\frac{1}{2y} + \frac{1}{y} = \frac{1}{6}; y = 9 \text{ days}$$

136. A and B working together can finish painting a house in 6 days. A working alone can finish it in 5 days less than B. How long will it take each of them to finish the work alone?

Solution:

$$\frac{1}{A} + \frac{1}{B} = \frac{1}{6} \text{ (Equation 1)}$$

$$A = B - 5 \text{ (Equation 2)}$$

Substitute (2) in (1):

$$\frac{1}{B-5} + \frac{1}{B} = \frac{1}{6} = \frac{2B-5}{B^2-5B}; B^2 - 17B + 30; B = 15 \text{ days \& } A = 15 - 5 = 10 \text{ days}$$

137. A and B can do a piece of work in 42 days, B and C in 31 days and C and A in 20 days. In how many days can all of them do the work together?

Solution:

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = \frac{1}{x}$$

$$\frac{1}{A} + \frac{1}{B} = \frac{1}{42} \text{ (Equation 1)}$$

$$\frac{1}{B} + \frac{1}{C} = \frac{1}{31} \text{ (Equation 2)}$$

$$\frac{1}{A} + \frac{1}{C} = \frac{1}{20} \text{ (Equation 3)}$$

Add three equations:

$$\left[\frac{1}{A} + \frac{1}{B} \right] + \left[\frac{1}{B} + \frac{1}{C} \right] + \left[\frac{1}{A} + \frac{1}{C} \right] = \frac{1}{42} + \frac{1}{31} + \frac{1}{20} = 0.106$$

$$\frac{2}{A} + \frac{2}{B} + \frac{2}{C} = 0.106$$

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = 0.053 = \frac{1}{x}; x = 18.87$$

138. Mike, Loui and Joy can mow the lawn in 4, 6, and 7 hours respectively. What fraction of the yard can they mow in 1 hour if they work together?

Solution:

Let x = the fraction of the lawn that can be mowed after one hour.

$$x = \left(\frac{1}{4} + \frac{1}{6} + \frac{1}{7} \right) (1) = \frac{42 + 28 + 24}{168} = \frac{47}{84}$$

139. A farmer can plow the farm in 8 days. After working for three days, his son joins him and together they plow the field 3 more days. How many days will it require for the son to plow the field alone?

Solution:

Let x = time for the farmer to plow the field

y = time for the son to plow the field.

$$\frac{1}{x}(3) + \left[\frac{1}{x} + \frac{1}{y}\right](3) = 1 \text{ (Equation 1)}$$

Substitute $x = 8$ in Equation (1)

$$\left[\frac{1}{8}(3) + \left[\frac{1}{8} + \frac{1}{y}\right](3) = 1\right] \left(\frac{1}{3}\right)$$

$$\frac{1}{8} + \frac{1}{8} + \frac{1}{y} = \frac{1}{3}; y = 12 \text{ days}$$

140. Crew no. 1 can finish installation of an antenna tower in 200man-hour while crew no. 2 can finish the same job in 300man-hour. How long will it take both crews to finish the same job, working together?

Solution:

Let x = the number of man-hours needed by crew number 1 and number 2 to finish the job.

$$\frac{1}{200} + \frac{1}{300} = \frac{1}{x}; x = 120 \text{man - hours}$$

141. On one job, two power shovels excavate 20,000 cubic meters of earth, the larger shovel working 40 hours and the smaller for 35 hours. On another job, they removed 40,000 cubic meters with the larger shovel working 70 hours and the smaller working 40 hours. How much earth can each remove in 1 hours working alone?

Solution:

Let x = the capacity of the larger shovel in m^3/hr .

y = the capacity of the smaller shovel in m^3/hr .

$$4x + 35y = 20,000 \text{ or } x = 500 - 0.875y \text{ (Equation 1)}$$

$$70x + 90y = 40,000 \text{ (Equation 2)}$$

Substitute (1) in (2):

$$70(500 - 0.875y) + 90y = 40,000; y = \frac{173.9\text{m}^3}{\text{hr.}} \text{ \& } x = \frac{347.8\text{m}^3}{\text{hr.}}$$

142. Ten liters of 25% salt solution and 15 liters of 35% salt solution are poured into a drum originally containing 30 liters of 10% salt solution. What is the percent concentration of salt in the mixture?

Solution:

$$\frac{25\%}{10} + \frac{35\%}{15} + \frac{10\%}{30} = \frac{x\%}{55}$$

$$0.25(10) + 0.35(15) + 0.1(30) = x(55); x = 19.55\%$$

143. A chemist of a distillery experimented on two alcohol solutions of different strength, 35% alcohol and 50% alcohol, respectively. How many cubic meters of each strength must he use in order to produce a mixture of 60 cubic meters that contain 40% alcohol?

Solution:

$$\frac{35\%}{x} + \frac{50\%}{60-x} = \frac{40\%}{60}$$

$$0.35(x) + 0.50(60 - x) = 0.4(60)$$

$$x = 40\text{m}^3 \text{ for the 35\% solution}$$

$$60 - x = 20\text{m}^3 \text{ for the 50\% solution}$$

144. A goldsmith has two alloys of gold, the first being 70% pure and the second being 60% pure. How many ounces of the 60% pure gold must be used to make 100 ounces of an alloy which will be 60% gold?

Solution:

$$\frac{70\%}{100-x} + \frac{60\%}{x} = \frac{66\%}{100}$$

$$0.70(100 - x) + 0.6(x) = 0.6(100); x = 40 \text{ ounces}$$

145. 2000kg of steel containing 8% nickel is to be made by mixing a steel containing 14% nickel with another containing 6% nickel. How much of each is needed?

Solution:

$$\frac{14\%}{x} + \frac{6\%}{2000 - x} = \frac{8\%}{2000}$$

$$0.14(x) + 0.06(2000 - x) = 0.08(2000);$$

$$x = 500\text{kg for the 14\% substance}$$

$$2000 - x = 1500\text{kg for the 6\% substance}$$

146. How much water must be evaporated from 10kg solution which has 4% salt to make a solution of 10% salt?

Solution:

$$\frac{4\%}{10} - \frac{0\%}{x} = \frac{10\%}{10 - x}$$

$$0.04(10) - 0(x) = 0.1(10 - x); x = 6\text{kg}$$

147. One number is less than the other. If their sum is 135, what are the numbers?

Solution:

Let x = tens' digit of the number

y = units' digit of the number

The two digit number is represented by: $10y + x$

148. Ten less than four times a certain number is 14. Determine the number.

Solution:

Let x = the first number

x - 5 = the second number

$$x + (x - 5) = 135;$$

$$x = 70 \text{ \& } x - 5 = 65$$

149. The sum of two numbers is 21 and one number is twice the other. Find the numbers.

Solution:

Let x = the first number

2x = the second number

$$x + 2x = 21$$

$$x = 7 \text{ \& } 2x = 14$$

150. If eight is added to the product of 9 and the numerical number, the sum is seventy-one. Find the unknown number.

Solution:

Let x = the number

$$9x + 8 = 71; x = 7$$

151. The product of $\frac{1}{4}$ and $\frac{1}{5}$ of a number is 500. What is the number?

Solution:

Let x = the number

$$\left[\frac{1}{4}(x)\right]\left[\frac{1}{5}(x)\right] = 500; x = 100$$

152. Find the product of two numbers such that twice the first added to the second equals 19 and three times the first is 21 more than the second.

Solution:

Let x = the first number

y = the second number

$$2x + y = 19 \text{ or } y = 19 - 2x \text{ (Equation 1)}$$

$$3x = y + 21 \text{ (Equation 2)}$$

Substitute (1) in (2):

$$3x = 19 - 2x + 21$$

$$x = 8 \text{ \& } y = 3$$

The product of the numbers is $8(3) = 24$

153. A jogger starts a course of a steady-rate of 8kph. Five minutes later, a second jogger starts the same course at 10kph. How long will it take the second jogger to catch the first?

Solution:

Let V_1 = Velocity of the first jogger

V_2 = Velocity of the second jogger

$$V_1 = \left(\frac{8\text{km}}{\text{hr}}\right)\left(\frac{1\text{hr}}{60\text{min}}\right) = \frac{8\text{km}}{60\text{min}}$$

$$V_2 = \frac{10\text{km}}{60\text{min}}$$

$$S_1 = S_2$$

$$V_1 t_1 = V_2 t_2$$

$$\frac{8}{10}(t_1) = \frac{10}{60}(t_1 - 5); t_1 = 25 \text{ minutes \& } t_2 = 25 - 5 = 20 \text{ minutes}$$

154. A boat man rows to a place 4.8 miles with the stream and back in 14 hours, but finds that he can row 14 miles with the stream in the same time as 3 miles against the stream. Find the rate of the stream. 184

Solution:

Let V_1 = Velocity of the boat man

V_2 = Velocity of the stream

$$\frac{4.8}{V_1 + V_2} + \frac{4.8}{V_1 - V_2} = 14$$

$$\frac{4.8V_1 - 4.8V_2 + 4.8V_1 + 4.8V_2}{V_1^2 - V_1V_2 + V_1V_2 - V_2^2} = 14$$

$$9.6V_1 = 14V_1^2 - 14V_2^2 \text{ (Equation 1)}$$

$$\frac{14}{V_1 + V_2} = \frac{3}{V_1 - V_2}; V_1 = 1.545V_2 \text{ (Equation 2)}$$

Substitute (2) in (1):

$$9.6(1.545V_2) = 14(1.545V_1)^2 - 14V_2^2$$

$$14.832V_2 = 19.418V_2^2; V_2 = 0.76\text{mph}$$

155. A man rows downstream at the rate of 5mph and upstream at the rate of 2mph. How far downstream should he go of he is to return in $\frac{7}{4}$ hours after leaving?

Solution:

Note: time = distance/ velocity

$$t_1 + t_2 = t_{\text{total}}$$

$$\frac{S}{5} + \frac{S}{2} = \frac{7}{4}; S = 2.5 \text{ miles}$$

156. An airplane flying with the wind, took 2 hours to travel 1000km and 2.5 hours in flying back. What is the wind velocity in kph?

Solution:

Let V_1 = velocity of the airplane

V_2 = velocity of the wind

$$V_1 + V_2 = \frac{1000}{2} = 500 \text{ (Equation 1)}$$

$$V_1 - V_2 = \frac{1000}{2.5} = 400 \text{ (Equation 2)}$$

Subtract (2) in (1)

$$(V_1 + V_2) - (V_1 - V_2) = 500 - 400; V_2 = 50 \text{ kph}$$

157. A boat travels downstream in $\frac{2}{3}$ of the time as it goes going upstream. If the velocity of the river's current is 8kph, determine the velocity of the boat in still water.

Solution:

Let V = velocity of the boat in still water

S_1 = distance traveled upstream

S_1 = distance traveled downstream

$$S_1 = S_2$$

$$(V - 8)(t) = (V + 8) \left[\frac{2}{3}t \right]$$

$$V - 8 = \frac{2}{3}V + \frac{16}{3}; V = 40 \text{ kph}$$

158. Two planes leave Manila for a southern city, a distance of 900km. Plane A travels at a ground speed of 90kph faster than the plane B. Plane A arrives in their destination 2 hours and 15 minutes ahead of plane B. What is the ground speed of plane A? 188

Solution:

Let V_1 = ground speed of plane A

V_2 = ground speed of plane B

$$S_1 = S_2$$

$$Vt = (V + 90)(t - 2.25)$$

$$2.25V - 90t + 202.5 = 0$$

$$\text{But } t = \frac{900}{V}$$

$$2.25V - 90 \left(\frac{900}{V} \right) + 202.5 = 0$$

Multiply both sides by V :

$$2.25V^2 + 202.5V - 81000 = 0; V = 240 \text{ kph}$$

159. On a certain trip, Edgar drive 231km in exactly the same time as Erwin drive 308km. if Erwin's rate exceeded that of Edgar by 13kph, determine the rate of Erwin. 190

Solution:

Let V = rate of Erwin

$V - 13$ = rate of Edgar

$$t_1 = t_2$$

$$\frac{231}{V - 13} = \frac{308}{V}; V = 52\text{kph}$$

160. The electric power which a transmission line can transmit is proportional to the product of its design voltage and current capacity, and inversely to the transmission distance. A 115 kilovolt line rated at 100 amperes can transmit 150 megawatts over 150km. How much power, in megawatts can a 230 kilovolt line rated at 150 amperes transmit over 100km?

Solution:

$$P = k \left(\frac{VI}{d} \right); \text{ when } P = 150\text{MW}, V = 115\text{volts}, I = 100\text{amperes and } d = 150\text{km}, k = ?$$

$$k = \frac{Pd}{VI} = \frac{(150\text{MW})(150\text{km})}{(115\text{V})(100\text{km})} = 1.956$$

When $V = 230\text{V}, I = 150\text{ampere}, d = 100\text{km}, P = ?$

$$P = (1.956) \left(\frac{(230\text{V})(150\text{amp})}{100\text{km}} \right) = 674.82\text{MegaWatts}$$

161. The time required for an elevator to lift a weight varies directly with the weight and the distance through which it is to be lifted and inversely as the power of the motor. If it takes 30 seconds for a 10hp motor to lift 100lb through 50ft, what size of motor is required to lift 800lb in 40 seconds through 40 ft.?

Solution:

$$t = k \left(\frac{WS}{P} \right); \text{ when } t = 30\text{s}, W = 100\text{lb}, S = 50\text{ft}, P = 10\text{hp}, k = ?$$

$$k = \frac{Pt}{WS} = \frac{(10\text{hp})(30\text{s})}{(100\text{lb})(50\text{ft})} = 0.06$$

When $t = 40\text{s}, W = 800\text{lb}$ and $S = 40\text{ft}$, what is P ?

$$P = k \left(\frac{WS}{t} \right) = (0.06) \left(\frac{(800\text{lb})(40\text{ft})}{40\text{s}} \right) = 48\text{hp}$$

162. The selling price of a TV set is double that of its cost. If the TV set was sold to a customer at a profit of 25% of the net cost, how much discount was given to the customer?

Solution:

Let x = net cost

$2x$ = Selling price

d = discount

$2x(1 - d)$ = new selling price

New selling price = Net cost + Gain

$$2x(1 - d) = x + 0.25x; d = 37.5\%$$

163. A group of AE examinees decided to hire a mathematics tutor from excel review center and planned to contribute equal amount for the tutor's fee. If there were 10 more examinees, each would have paid ₱2 less. However, if there were 5 less examinees, each would have paid ₱2 more. How many examinees are there in the group?

Solution:

Let x = number of examinees

y = tutor's fee

x/y = original fee shared per examinee

If there were 10 more examinees who will join,

$$y = (x + 10) \left(\frac{y}{x} - 2 \right); y = 0.2x^2 + 2x \text{ (Equation 1)}$$

IF there were 5 examinees who will back out,

$$y = (x - 5) \left(\frac{y}{x} + 2 \right); y = 0.4x^2 + 2x \text{ (Equation 2)}$$

Equate (1) & (2):

$$0.2x^2 + 2x = 0.4x^2 + 2x; x = 20 \text{ examinees}$$

164. A bookstore purchased a best-selling price book at ₱200 per copy. At what price should this book be sold so that, giving 20% discount, the profit is 30%?

Solution:

Let x = selling price without discount

$0.8x$ = new selling price (with discount)

Profit = Income – Expenses

$$0.3(0.8x) = 0.8x - 200; x = \text{₱}357.14$$

165. Jojo bought a second hand Betamax VCR and then sold it to Rudy at a profit of 40%. Rudy then sold the VCR to Noel at a profit of 20%. If Noel paid ₱2,856 more than it cost to Jojo, how much did Jojo paid for the unit?

Solution:

Let x = price Jojo paid for the VCR

$1.4x$ = price Rudy paid for the VCR

$1.2(1.4x)$ = price Noel paid for the VCR

$$1.2(1.4x) = x + 2856; x = \text{₱}4,200$$

166. In a certain community of 1,200 people, 60% are literate. Of the males, 50% are literate and the females 70% are literate. What is the female population?

Solution:

Let x = the number of men in the population

y = number of female in the population

$$x + y = 1200; x = 1200 - y \text{ (Equation 1)}$$

$$0.5x + 0.7y = 0.6(1200); \text{ (Equation 2)}$$

Substitute (1) in (2):

$$0.5(1200 - y) + 0.7y = 720; y = 600 \text{ females}$$

167. A merchant has three items on sale, namely a radio for ₱50, a clock for ₱30 and a flashlight for ₱1. At the end of the day, he sold a total of 100 of the three items and has taken exactly ₱1,000 on the total sales. How many radios did him sale?

Solution:

Let x = the number of radios sold out

y = number of clocks sold out

z = number of flashlight sold out

$$x + y + z = 100 \text{ (Equation 1)}$$

$$50x + 30y + z = 1000 \text{ (Equation 2)}$$

Subtract (2) by (1) and you will get:

$$49x + 29y = 900$$

Assume $x = 16$:

$$49(16) + 29y = 900; y = 4$$

Substitute $x = 20$ and $y = 4$ in (1) and you get:

$$z = 80$$

Substitute both values of x , y and z in (2) and you get:

$$50(16) + 30(4) + 80 = 1000$$

$$1000 = 1000 : \text{thus, } x = 16 \text{ radios}$$

168. The arithmetic mean of a and b is?

Solution:

Note: arithmetic mean is the same term as average.

Thus, the arithmetic mean of a and b is, $\frac{a+b}{2}$

169. The sum of the arithmetic mean between 34 and 42 is?

Solution:

$$34, a_2, a_3, a_4, 42$$

$$a_5 = a_1 + 4d$$

$$42 = 34 + 4d; d = 2$$

$$\text{Thus, } a_2 = 36, a_3 = 38 \text{ and } a_4 = 40$$

170. If the first term of the arithmetic progression is 25 and the fourth term is 13, what is the third term?

Solution:

$$a_1 = 25; a_4 = 13$$

$$a_4 = a_1 + 3d$$

$$13 = 25 + 3d; d = -4$$

$$a_3 = a_1 + 2d$$

$$a_3 = 25 + 2(-4) = 17$$

171. Find the 30th term of the arithmetic progression 4, 7, 10...

Solution:

$$a_1 = 4; a_2 = 7; a_3 = 10$$

$$\text{by inspection: } d = 3$$

$$a_{30} = a_1 + 29d$$

$$a_{30} = 4 + 29(3)$$

$$a_{30} = 91$$

172. How many items of the progression 3, 5, 7... must be taken in order that their sum will be 2600?

Solution:

$$a_1 = 3; a_2 = 5; a_3 = 7$$

$$\text{by inspection: } d = 2$$

$$S = \frac{n}{2}[2a_1 + (n-1)d]$$

$$2600 = \frac{n}{2}[2(3) + (n-1)2] = \frac{n}{2}[6 + 2n - 2] = \frac{n}{2}[4 + 2n]$$

$$2600 = 2n + n^2; n = 50 \text{ numbers}$$

173. In a pile of logs, each layer contains one more log than the layer above and the top contains just one log. If there are 105 logs in the pile, how many layers are there?

Solution:

$$a_1 = 1; d = 1; S = 105$$

$$S = \frac{n}{2}[2a_1 + (n-1)d]$$

$$105 = \frac{n}{2}[2(1) + (n-1)(1)] = \frac{n}{2}[2 + n - 1] = \frac{n}{2}[1 + n]$$

$$210 = n + n^2; n = 14 \text{ layers of log}$$

174. What is the sum of the progression 4, 9, 14, 19 ... up to the 20th term?

Solution:

$$a_1 = 4; a_2 = 9; a_3 = 14; a_4 = 19; n = 20$$

By inspection: $d = 5$

$$S = \frac{n}{2}[2a_1 + (n-1)d] = \frac{20}{2}[2(4) + 19(5)] = 1030$$

175. A stack of bricks in the bottom layer, 58 bricks in the second layer, 55 bricks in the third layer, and so on until there are 10 bricks in the last layer. How many bricks are there all together?

Solution:

$$a_1 = 61; a_2 = 58; a_3 = 55; a_n = 10$$

By inspection, $d = -3$

$$a_n = a_1 + (n-1)d$$

$$10 = 61 + (n-1)(-3); n = 18$$

$$S = \frac{n}{2}[2a_1 + (n-1)d] = \frac{18}{2}[2(61) + (18-1)(-3)]$$

$$S = 639 \text{ logs}$$

176. Determine the sum of the progression if there are 7 arithmetic mean between 3 and 35.

Solution:

$$3, a_2, a_3, a_4, a_5, a_6, a_7, a_8, 35$$

by inspection: $a_1 = 3; a_n = 35; n = 7 + 2 = 9$

$$S = \frac{n}{2}(a_1 + a_n) = \frac{9}{2}(3 + 35)$$

$$S = 171$$

177. If the sum is 220 and the first term is 10, find the common difference if the last term is 30.

Solution:

$$S = 220; a_1 = 10; a_n = 30$$

$$S = \frac{n}{2}(a_1 + a_n)$$

$$220 = \frac{n}{2}(10 + 30); n = 11$$

$$a_{11} = a_1 + 10d$$

$$30 = 10 + 10d$$

$$d = 2$$

178. Once a month, a man puts some money into the cookie jar. Each month he puts 50 centavos more into the jar than the month before. After 12 years, he counted his money, he had ₱5,436. How much did he put in the jar in the last month?

Solution:

$$d = 0.50; n = 12(12) = 144$$

$$S = \frac{n}{2}[2a_1 + (n-1)d]$$

$$5436 = \frac{144}{2}[2a_1 + (144-1)(0.50)]; a_1 = 2$$

$$a_{144} = a_1 + 143d = 2 + 143(0.5) = 73.50$$

179. When all odd number from 1 to 101 are added, the result is?

Solution:

$$a_1 = 1; d = 2; n = 51$$

$$S = \frac{n}{2}[2a_1 + (n-1)d] = \frac{51}{2}[2(1) + (51-1)(2)]$$

$$S = 1300$$

180. To conserve energy due to the present energy crisis, the Meralco tried to re-adjust their charges to electrical energy users who consume more than 2000kW-hrs. For the first 100kW-hr, they charged 40 centavos and increasing at a constant rate more than the preceding one until the fifth 100kW-hr, the charge is 76 centavos. How much is the average charge for the electrical energy per 100kW-hr.?

Solution:

$$a_1 = 40; a_5 = 76$$

$$a_5 = a_1 + 4d$$

$$75 = 40 + 4d; d = 9$$

$$a_2 = 40 + 9 = 49$$

$$a_3 = 49 + 9 = 58$$

$$a_4 = 58 + 9 = 67$$

$$\text{Average} = \frac{40 + 49 + 58 + 67 + 76}{5} = 58 \text{ centavos}$$

181. The third term of a harmonic progression is 15 and the 9th term is 6. Find the 11th term.

Solution:

$$h_3 = 15; a_3 = \frac{1}{15}; h_9 = 6; a_9 = \frac{1}{6}$$

$$a_3 = a_1 + 2d$$

$$\frac{1}{15} = a_1 + 2d; a_1 = \frac{1}{15} - 2d \text{ (Equation 1)}$$

$$a_9 = a_1 + 8d$$

$$\frac{1}{6} = a_1 + 8d; a_1 = \frac{1}{6} - 8d \text{ (Equation 2)}$$

Substitute (1) in (2):

$$\frac{1}{6} = \frac{1}{15} - 2d + 8d; d = \frac{1}{60}$$

Substitute d value in (1):

$$a_1 = \frac{1}{15} - 2d = \frac{1}{15} - 2\left(\frac{1}{60}\right) = \frac{1}{30}$$

$$a_{11} = a_1 + 10d = \frac{1}{30} + 2\left(\frac{1}{60}\right) = \frac{1}{5}$$

$$\text{Thus, } h_{11} = \frac{1}{a_{11}} = \frac{1}{\frac{1}{5}} = 5$$

182. Find the fourth term of the progression $\frac{1}{2}, 0.2, 0.125, \dots$

Solution:

$$h_1 = \frac{1}{2}; h_2 = \frac{1}{5}; h_3 = \frac{1}{8}; a_1 = 2; a_2 = 5; a_3 = 8$$

By inspection: $d = 3$

$$a_4 = a_1 + 3d = 2 + 3(3) = 11$$

$$\text{Thus, } h_4 = \frac{1}{a_4} = \frac{1}{11}$$

183. Find the 9th term of the harmonic progression 3, 2, $\frac{3}{2}, \dots$

Solution:

$$h_1 = 3; h_2 = 2; a_1 = \frac{1}{3}; a_2 = \frac{1}{2}$$

$$d = a_2 - a_1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$a_9 = a_1 + 8d = \frac{1}{3} + 8\left(\frac{1}{6}\right) = \frac{5}{3}$$

$$h_9 = \frac{1}{a_9} = \frac{1}{\frac{5}{3}} = \frac{3}{5}$$

184. Find the sum of 4 geometric means between 160 and 5.

Solution:

$$a_1 = 160; a_6 = 5$$

$$a_6 = a_1 r^5$$

$$5 = 160r^5; r = 0.5$$

Thus,

$$a_2 = 160(0.5) = 80$$

$$a_3 = 80(0.5) = 40$$

$$a_4 = 40(0.5) = 20$$

$$a_5 = 20(0.5) = 10$$

$$\text{Sum} = 80 + 40 + 20 + 10 = 150$$

185. The fourth term of a Geometric progression is 216 and the 6th term is 1944. Find the 8th term.

Solution:

$$a_4 = 216; a_6 = 1944$$

$$a_4 = a_1 r^3$$

$$216 = a_1 r^3 \text{ (Equation 1)}$$

$$a_6 = a_1 r^5$$

$$1944 = a_1 r^5 \text{ (Equation 2)}$$

Divide (2) by (1):

$$\frac{a_1 r^5}{a_1 r^3} = \frac{1944}{216}; r = 3$$

Substitute r value in (1)

$$216 = a_1 (3)^3; a_1 = 8$$

$$a_8 = a_1 r^7 = (8)(3)^7 = 17496$$

186. Determine x so that: x, 2x + 7, 10x - 7 will be geometric progression.

Solution:

$$a_1 = x; a_2 = 2x + 7; a_3 = 10x - 7;$$

$$r = \frac{a_2}{a_1} = \frac{a_3}{a_2}$$

$$\frac{2x + 7}{x} = \frac{10x - 7}{2x + 7}; 6x^2 - 35x - 49 = 0$$

$$x_1 = 7; x_2 = -\frac{7}{6}$$

187. If one third of the air tank is removed by each stroke of an air pump, what fractional part of the total air is removed in 6 strokes?

Solution:

Let x = Total volume of air in the tank

y = Total volume removed from the tank after the 6th stroke

$$\text{Volume left after 1st stroke} = x - \frac{1}{3}x = \frac{2}{3}x$$

$$\text{Volume left after 1st stroke} = \frac{2}{3}x - \frac{1}{3}\left(\frac{2}{3}x\right) = \frac{4}{9}x$$

$$\text{Volume left after 1st stroke} = \frac{4}{9}x - \frac{1}{3}\left(\frac{4}{9}x\right) = \frac{8}{27}x$$

By inspection, the volume left after each stroke forms a Geometric progression whose common

$$\text{ratio } r = \frac{\frac{8}{27}}{\frac{4}{9}} = \frac{2}{3}$$

Solving for the volume left after the 6th stroke:

$$a_6 = a_1 r^5 = \left[\frac{2}{3}x\right] \left[\frac{2}{3}\right]^5 = 0.08779x$$

Thus, the total volume left after the 6th stroke:

$$y = x - 0.08779x = 0.9122x$$

188. A product has a current selling of ₱325. If it's selling price is expected to decline at the rate of 10% per annum because of obsolescence, what will be its selling price four year hence?

Solution:

$$r = 0.9$$

$$a_1 = 325(0.9) = 292.5$$

$$a_4 = a_1 r^3 = (292.5)(0.9)^3 = \text{₱}213.33$$

189. The sum of the first 10 terms of a geometric progression 2, 4, 8 ... is?

Solution:

$$a_1 = 2; a_2 = 4; a_3 = 8; n = 10$$

By inspection: $r = 2$

$$S = \frac{a_1(r^n - 1)}{r - 1} = \frac{2(2^{10} - 1)}{2 - 1} = 2046$$

190. A rubber ball is made to fall from a height of 50ft and is observed to rebound $\frac{2}{3}$ of the distance it falls. How far will the ball travel before coming to rest if the ball continues to fall in this manner?

Solution:

$$a_1 = 50 \left[\frac{2}{3}\right] = \frac{100}{3}; n = \infty$$

$$S = \frac{a_1}{1 - r} = \frac{\frac{100}{3}}{1 - \frac{2}{3}} = 100$$

Let D = Total distance traveled by ball

$$D = 50 + 2S$$

$$D = 250\text{ft}$$

191. Find the sum of the infinite geometric progression 6, -2, $\frac{2}{3}$, ...

Solution:

$$a_1 = 6; a_2 = -2$$

$$r = \frac{a_2}{a_1} = -\frac{2}{6} = -\frac{1}{3}$$

$$S = \frac{a_1}{1 - r} = \frac{6}{1 - \left(-\frac{1}{3}\right)} = \frac{9}{2}$$

192. Find the sum of 1, $-\frac{1}{5}$, $\frac{1}{25}$, ... 250

Solution:

$$a_1 = 1; a_2 = -\frac{1}{5}; a_3 = \frac{1}{25}$$

By inspection: $r = -\frac{1}{5}$

$$S = \frac{a_1}{1-r} = \frac{1}{1 - \left(-\frac{1}{5}\right)} = \frac{5}{6}$$

193. In a class of 40 students, 27 like calculus and 25 like chemistry. How many like both calculus and chemistry?

Solution:

Let x = number of students who like both subjects.

$$(27 - x) + x + (25 - x) = 40; x = 12 \text{ students}$$

194. A survey of 100 persons revealed that 72 of them had eaten at restaurant P and that 52 of them had eaten at restaurant Q. Which of the following could not be the number of persons in the surveyed group who had eaten at both P and Q?

Solution:

Let x = number of persons who have eaten on both restaurants

$$(72 - x) + x + (52 - x) = 100; x = 24 \text{ persons}$$

195. The probability for the ABE board examinees from a certain school to pass the subjects mathematics is $\frac{3}{7}$ and for the subject Agricultural machinery is $\frac{5}{7}$. If none of the examinees fails both subject and there are 4 examinees who pass both subjects, find the number of examinees from that school who took the examinations.

Solution:

Let x = number of examinees who took the examination

$$x = \left[\frac{3}{7}x - 4\right] + 4 + \left[\frac{5}{7}x - 4\right]; x = 28 \text{ examinees}$$

196. In a commercial survey involving 1000 persons on brand reference, 120 were found to prefer brand x only, 200 prefer brand y only, 150 prefer brand z only, 370 prefer either brand x or y but not z, 450 prefer brand y or z but not x and 370 prefer either brand z or x but not y. how many persons have no brand reference, satisfied with any of the three brands?

Solution:

Let x = number of persons who have no brand reference

$$1000 = x + 120 + 50 + 200 + 100 + 150 + 100; x = 280 \text{ persons}$$

197. If 15 people won prizes in the state lottery (assuming that there are no ties), how many ways can these 15 people win first, second, third fourth and 5th prizes?

Solution:

15	14	13	12	11
1 st	2 nd	3 rd	4 th	5 th
price	price	price	price	price

$$N = 15(14)(13)(12)(11)$$

$$N = 360,360 \text{ ways}$$

198. How many 4 digit numbers can be formed without repeating any digit from the following digits: 5, 4, 3 and 2?

Solution:

5	4	3	2
1 st	2 nd	3 rd	4 th
digit	digit	digit	digit

$$N = 5(4)(3)(2)$$

$$N = 120 \text{ numbers}$$

199. How many permutations are there if the letter PNRCS are taken six at a time?

Solution:

$$nPr = n!$$

$$6P6 = 6! = 720 \text{ ways}$$

200. In how many ways can 6 distinct books arranged in a bookshelf?

Solution:

Same solution and answer in #199

201. What is the permutations of the letters in the word BANANA?

Solution:

Note: "BANANA"

Number of A's = 3

Numbers of N's = 2

$$nPr, q = \frac{n!}{p! q!} = \frac{6!}{3! 2!} = 60 \text{ ways}$$

202. A PSME unit has 10 ME's 8 PME's and 6 CPM's. If a committee of 3 members, one from each group is to be formed, how many such committees can be formed?

Solution:

10	8	6
M ₁	M ₂	M ₃

$$N = 10(8)(6)$$

$$N = 480 \text{ ways}$$

203. Four colored flags can be hung in a row to make signal. How many signals can be if a signals consists of the display of one or more flags?

Solution:

With one flag:

$$N_1 = 4 \text{ signals}$$

4

With two flags:

$$N_2 = 4(3) = 12 \text{ signals}$$

4	3
---	---

With three flags:

$$N_3 = 4(3)(2) = 24 \text{ signals}$$

4	3	2
---	---	---

With four flags:

$$N_4 = 4(3)(2)(1) = 24 \text{ signals}$$

4	3	2	1
---	---	---	---

$$N = 4 + 12 + 24 + 24 = 64 \text{ signals}$$

204. In how many ways can 4 boys and 4 girls be seated alternately in a row of 8 seats?

Solution:

Number of ways the 4 boys can be arranged = 4!

Number of ways the 4 girls can be arranged = 4!

$$N = (4!)(4!) = 1152 \text{ ways}$$

205. There are four ball of four different colors. Two balls are taken at a time and arranged in a definite order. For example, if a white and a red ball are taken, one definite arrangement is white first, red second and another arrangement is red first, white second. How many such arrangement are possible?

Solution:

$$nPr = \frac{n!}{(n-r)!}$$

$$4P2 = \frac{4!}{(4-2)!} = 12 \text{ arrangements}$$

206. There are four ball of four different colors. Two balls are taken at a time and arranged in any way. How many such combinations are possible?

Solution:

Note: Since the objects taken can be arranged in any way, then the said arrangement is under the principles of “combination”

$$nPr = \frac{n!}{(n-r)!}$$

$$4P2 = \frac{4!}{(4-2)!2!} = 6 \text{ combinations}$$

207. How many 6 combinations can be generated from the numbers from 1 to 42 inclusive, without repetition and with no regards to the order of the numbers?

Solution:

$$42P6 = \frac{42!}{(42-6)!6!} = 5,245,786 \text{ combinations}$$

208. In how many ways can you invite one or more of your five friends in a party?

Solution:

$${}^5C_{1,2,3,\dots} = 2^5 - 1 = 31 \text{ combinations}$$

209. There are 50 tickets in a lottery in which there is a first and second prize. What is the probability of a man drawing a prize if he owns 5 tickets?

Solution:

P = probability of the man to win a prize

P = number of tickets he bought x probability of winning the lottery

$$P = 5 \left[\frac{2}{50} \right] = \frac{1}{5}$$

210. Roll a pair of dice. What is the probability that the sum of two numbers is 11?

Solution:

Number of trials with a sum of 11 = 2 trials

Total number of trials = $6^n = (6)^2 = 36$ trials

$$P = \frac{\text{Number of successful trials}}{\text{total number of trials}} = \frac{2}{36} = \frac{1}{18}$$

211. Determine the probability of drawing either king or a diamond in a single draw from a pack of 52 playing cards.

Solution:

Let P_k = probability of drawing a king

P_D = probability of drawing a diamond

$P_{k \& D}$ = probability of drawing a king at the same time a diamond

$$P_k = \frac{4}{52}; P_D = \frac{13}{52}; P_{k \text{ or } D} = \frac{1}{52}$$

$$P_{k \text{ or } D} = P_k + P_D - P_{k \& D}$$

$$P_{k \text{ or } D} = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{4}{13}$$

212. A card is drawn from a deck of 52 playing cards. Find the probability of drawing a king or a red card.

Solution:

Let P_k = probability of drawing a king

P_R = probability of drawing a red card

$P_{k \& R}$ = probability of drawing a king at the same time a red card

$$P_k = \frac{4}{52}; P_R = \frac{26}{52}; P_{k \text{ or } R} = \frac{2}{52}$$

$$P_{k \text{ or } R} = P_k + P_R - P_{k \& R}$$

$$P_{k \text{ or } D} = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = 0.5385$$

213. A coin is tossed 3 times. What is the probability of getting 3 tails up?

Solution:

$$P = nCr p^r q^{n-r}$$

Where:

P = Probability of getting 3 tails out from 3 trials

p = Probability of getting a tail from a single trial

q = Probability of not getting a tail from a single trial

n = number of trials

r = number of successful trials

$$p = \frac{1}{2}; q = \frac{1}{2}; n = 3; r = 3$$

Substitute:

$$P_{3T} = {}^3C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^0 = \frac{3!}{(3-3)! 3!} \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

214. The probability of getting a credit in an examination is 1/3. If three students are selected at random, what is the probability that at least one of them got a credit?

Solution:

Note: Probability of getting a credit card = 1/3

Probability of not getting any credit = 2/3

Let P_1 = Probability that only one student gets a credit

$$P_1 = {}^3C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^2 = \frac{3!}{(3-1)! 1!} \left(\frac{4}{27}\right) = \frac{12}{27}$$

P_2 = Probability that exactly 2 students gets a credit

$$P_2 = {}^3C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^1 = \frac{3!}{(3-2)! 2!} \left(\frac{2}{27}\right) = \frac{6}{27}$$

P_3 = Probability that all three student gets a credit

$$P_2 = {}^3C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^0 = (1) \left(\frac{1}{27}\right) = \frac{1}{27}$$

Probability that at least one student gets a card

$$P = P_1 + P_2 + P_3 = \frac{12}{27} + \frac{6}{27} + \frac{1}{27}$$

$$P = \frac{19}{27}$$

215. In the ABE Board Examinations, the probability that an examinee will pass each subjects is 0.8. What is the probability that an examinee will pass at least two subjects out of the three board subject?

Solution:

Note: Probability of getting a passing score in each subject is 0.8

Probability of failing in any of the three subjects is 0.2

Let P_1 = Probability of passing exactly two subject

$$P_1 = {}^3C_2 (0.8)^2 (0.2)^1 = \frac{3!}{(3-2)! 2!} = 0.384$$

P_1 = Probability of passing all the three subjects

$$P_2 = {}^3C_3 (0.8)^3 (0.2)^0 = (1)(0.512) = 0.512$$

P = Probability of passing at least two subjects

$$P = P_1 + P_2 = 0.384 + 0.512$$

$$P = 0.896 \text{ or } 89.6\%$$

216. From a box containing 6 red balls, 8 white balls and 10 blue balls, one ball is drawn at random. Determine the probability that it is red or white.

Solution:

Let P = Probability of getting a red or a white ball from the box

$$P = \frac{\text{Number of red or white ball}}{\text{total number of trials}} = \frac{14}{24} = \frac{7}{12}$$

217. From a bag containing 4 black balls and 5 white balls, two balls are drawn one at a time. Find the probability that both balls are white. Assume that the first ball is returned before the second ball is drawn.

Solution:

Let P_1 = probability of drawing a white ball in the first draw

$$P_1 = \frac{5}{9}$$

P_2 = probability of drawing a white ball in the second draw

Note: the first ball was returned in the bag before the 2nd ball was drawn

$$P_2 = \frac{5}{9}$$

P = Probability that both balls drawn are all white

$$P = P_1 \times P_2 = \frac{5}{9} \times \frac{5}{9} = \frac{25}{81}$$

218. A group of 3 people enters a theater after the lights had dimmed. They are shown to the correct group of 3 seats by the usher. Each person holds a number stub. What is the probability that each is in the correct seat according to the numbers on seat and stub? 304

Solution:

Probability that A is correct = $1/3$, assuming he is to sit down first

Probability that B is correct = $1/2$, assuming he is to sit down after A

Probability that C is correct = 1 , assuming he is the last to sit down

Let P = Probability that A, B and C are correct

$$P = \frac{1}{3} \times \frac{1}{2} \times 1 = \frac{1}{6}$$

Alternate solution:

Total number of possible arrangement = $3! = 6$ arrangements

$$P = \frac{\text{Correct arrangement}}{\text{total number of arrangement}} = \frac{1}{6}$$

219. The angle subtended by an arc is 24° . If the radius of the circle is 45cm, find the length of the arc.

Solution:

$$C = r\theta = 45\text{cm} \left[24^\circ \times \frac{2\pi\text{rad}}{360^\circ} \right] = 18.85\text{cm (direct multiplication only)}$$

220. Each angle of a regular dodecagon is equal to

Solution:

Since, dodecagon has 12 sides,

$$\theta = \frac{(n-2)(180^\circ)}{n} = \frac{(12-2)(180)}{12} = 150^\circ$$

221. How many sides has a polygon if the sum of the interior angles is 1080° ?

Solution:

$$S = (n - 2)(180)$$

$$1080 = (n - 2)(180)$$

$n = 8$ sides

222. How many sides are in a polygon if each interior angles is 165 degrees?

Solution:

$$\theta = \frac{(n - 2)(180)}{n}$$

$$160^\circ = \frac{(n - 2)(180)}{n}$$

$n = 24$ sides

223. Given triangle, $C = 100^\circ$, $a = 15\text{m}$, $b = 20\text{m}$. Find c

Solution:

Law of cosines

$$c = \sqrt{a^2 + b^2 - 2ab\cos C} = \sqrt{(15\text{m})^2 + (20\text{m})^2 - 2(15)(20)\cos 100} = 27\text{m}$$

224. In triangle ABC, angle $A = 45$ degree and $C = 70$ degrees. The side opposite angle C is 40m long. What is the length of the side opposite angle A ?

Solution:

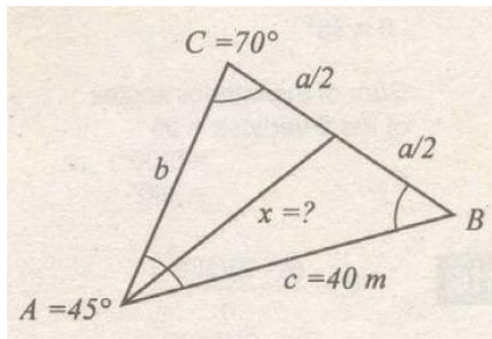
Sine Law

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$a = c \left[\frac{\sin A}{\sin C} \right] = 40\text{m} \left[\frac{\sin 45}{\sin 70} \right] = 30.1\text{m}$$

225. In triangle ABC, angle $C = 70$ degrees, $A = 45$ degrees and $AB = 40\text{m}$. What is the length of the median drawn from vertex A to BC ?

Solution:



$$\frac{\sin C}{c} = \frac{\sin A}{a}$$

$$\frac{\sin 70}{40\text{m}} = \frac{\sin 45}{a}; a = 30\text{m} \text{ \& } \frac{a}{2} = 15$$

$$A + B + C = 180^\circ$$

$$45^\circ + B + 70^\circ = 180; B = 65^\circ$$

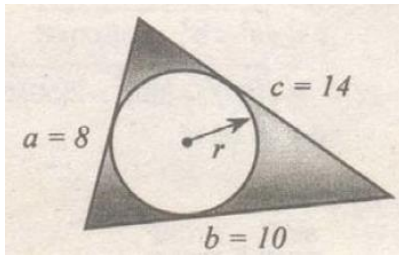
By cosine Law:

$$x = \sqrt{c^2 + a^2 - 2cac\cos B}$$

$$x = \sqrt{(40\text{m})^2 + (15)^2 - 2(40\text{m})(15)\cos 65^\circ} = 36.3\text{m}$$

226. The sides of a triangle are 8cm, 10cm and 14cm. determine the radius of the inscribed circle.

Solution:



$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$s = \frac{a+b+c}{2} = \frac{8\text{cm} + 10\text{cm} + 14\text{cm}}{2} = 16\text{cm}$$

$$A = \sqrt{(16\text{cm})(16-8)(16-10)(16-14)} = 39.19\text{cm}^2$$

$$A = rs$$

$$r = \frac{A}{s} = \frac{39.19}{16} = 2.45\text{cm}$$

227. What is the radius of the circle circumscribing an isosceles right triangle having an area of 162cm^2 ?

Solution:

Note: Since an isosceles right triangle, side "a" and "b" must be equal and side "c" is equal to the diameter of the circumscribing circle.

$$A_T = \frac{1}{2}ab = \frac{1}{2}a^2$$

$$162\text{cm}^2 = \frac{a^2}{2}; a = 18\text{cm}$$

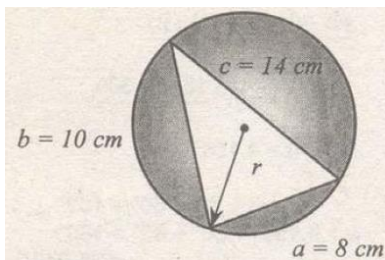
Using the Pythagorean Theorem:

$$c = \sqrt{a^2 + b^2} = \sqrt{18^2 + 18^2} = 25.46\text{cm}$$

$$\text{radius} = \frac{25.46\text{cm}}{2} = 12.73\text{cm}$$

228. The sides of a triangle are 8cm, 10cm and 14cm. determine the radius of the circumscribing circle.

Solution:



$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$s = \frac{a+b+c}{2} = \frac{8+10+14}{2} = 16\text{cm}$$

$$A = \sqrt{16(16-8)(16-10)(16-14)} = 39.19\text{cm}^2$$

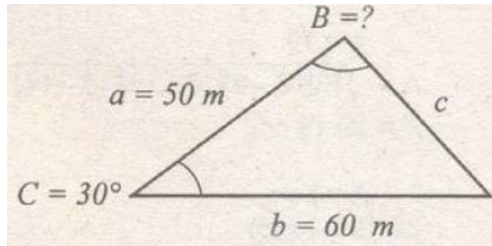
$$A = \frac{abc}{4r}$$

$$39.19\text{cm}^2 = \frac{(8\text{cm})(10\text{cm})(14\text{cm})}{4r}$$

$$r = 7.14\text{cm}$$

229. Two sides of the triangle are 50m and 60m long. The angle included between these sides is 30° . What is the interior angle opposite the longest side?

Solution:



Using Law of Cosines, we get the side c length

$$c = \sqrt{a^2 + b^2 - 2ab\cos\theta} = 30.06\text{m}$$

Those, the longest side is side b and we are required to find its opposite interior angle B.

Use the Law of Sine to find B.

$$\frac{\sin B}{60} = \frac{\sin 30}{30.06}$$

$$B = 86.38^\circ$$

230. The area of a circle is 89.42in^2 . What is its circumference?

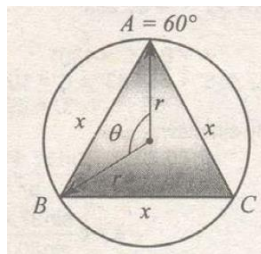
Solution:

$$A = \pi r^2; r = 5.335\text{in}$$

$$C = 2\pi r = 2\pi(5.335\text{in}) = 33.52\text{in}$$

231. The area of a circle circumscribing about an equilateral triangle is 254.47m^2 . What is the area of the triangle in m^2 ?

Solution:



$$A = \pi r^2; r = 9\text{m}$$

Note: An equilateral triangle,

$$\angle A = \angle B = \angle C = 60^\circ$$

$$\theta = \frac{360^\circ}{3} = 120^\circ$$

By cosine law:

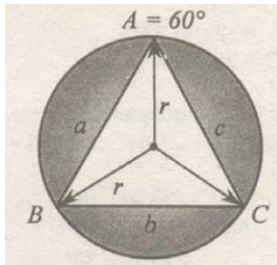
$$x = \sqrt{r^2 + r^2 - 2rr\cos\theta} = \sqrt{9^2 + 9^2 - 2(9)(9)\cos 120}$$

$$x = 15.588\text{m}$$

$$A_T = \frac{1}{2}bh\sin A = \frac{1}{2}xx\sin A = \frac{1}{2}(15.588)^2\sin 60 = 105.24\text{m}^2$$

232. What is the area in cm^2 of the circle circumscribed about an equilateral triangle with a side 10cm long?

Solution:



Note: An equilateral always has interior angle of 60° in every side.

And $\angle A = \angle B = \angle C = 60$

$$A_T = \frac{1}{2}bh\sin A = \frac{1}{2}(10\text{cm})^2\sin 60 = 43.3\text{cm}^2$$

$$A = \frac{abc}{4r}$$

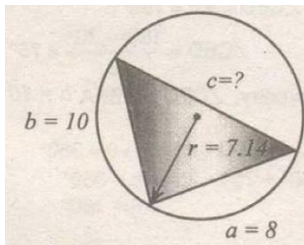
$$43.3\text{cm}^2 = \frac{(10\text{cm})(10\text{cm})(10\text{cm})}{4r}$$

$$r = 5.774\text{cm}$$

$$A_c = \pi r^2 = \pi(5.774\text{cm})^2 = 104.74\text{cm}^2$$

233. The area of a triangle inscribed in a circle is 39.19cm^2 and the radius of the circumscribed circle is 7.14cm . If the two sides of the triangle are 8cm and 10cm , find the third side.

Solution:



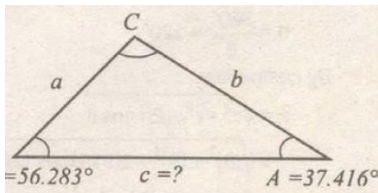
$$A = \frac{abc}{4r}$$

$$39.19\text{cm}^2 = \frac{(8\text{cm})(10\text{cm})c}{4(7.14\text{cm})}$$

$$c = 14\text{cm}$$

234. The area of a triangle is 8346m^2 and two of its interior angles are $37^\circ 25'$ and $56^\circ 17'$. What is the length of the longest side?

Solution:



$$A = 37^\circ 25' = 37.416^\circ$$

$$B = 56^\circ 17' = 56.283^\circ$$

$$\text{Total angle} = A + B + C = 180$$

$$37.416 + 56.283 + C = 180; C = 86.301^\circ$$

By sine Law:

$$\frac{\sin 86.301}{c} = \frac{\sin 37.416}{a}; a = c \left[\frac{\sin 37.416}{\sin 86.301} \right]$$

$$a = 0.609c$$

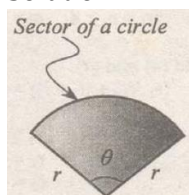
$$A = \frac{1}{2} ac \sin B = \frac{1}{2} (0.609c)(c) \sin 56.283$$

$$8346 \text{ m}^2 = \frac{1}{2} (0.609c)(c) \sin 56.283$$

$$c = 181.5 \text{ m}$$

235. The angle of a sector is 30° and the radius is 15cm. What is the area of the sector in cm^2 ?

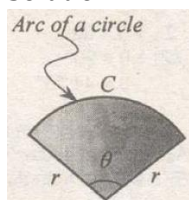
Solution:



$$A = \frac{1}{2} r^2 \theta = \frac{1}{2} (15 \text{ cm})^2 \left[30^\circ * \frac{2\pi \text{ rad}}{360} \right] = 58.9 \text{ cm}^2$$

236. The arc of a sector is 9 units and its radius is 3 units. What is the area of a sector in square units?

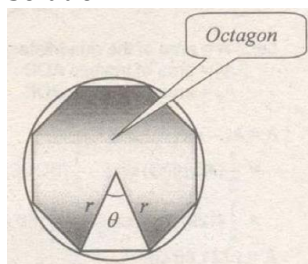
Solution:



$$A = \frac{1}{2} rC = \frac{1}{2} (3 \text{ units})(9 \text{ units}) = 13.5 \text{ square units}$$

237. Find the area in cm^2 of a regular octagon inscribed in a circle of radius 10cm?

Solution:

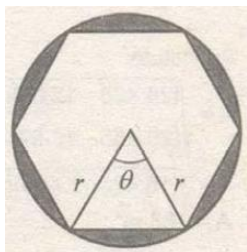


$$A = 8A_T = 8 \left[\frac{1}{2} r^2 \sin \theta \right] = 8 \left[\frac{1}{2} (10 \text{ cm})^2 \sin \left(\frac{360}{8} \right) \right] = 283 \text{ cm}^2$$

Note: If it is a Pentagon it will be $5A_T$, if Hexagon is $6A_T$

238. A regular Hexagon is inscribed in a circle whose diameter is 20m. Find the area of the 6 segments of the circle formed by the sides of the hexagon.

Solution:



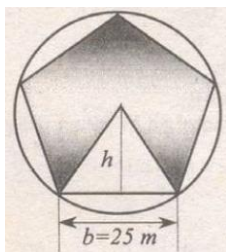
$$A_H = 6A_T = 6 \left(\frac{1}{2} r r \sin \theta \right) = 6 \left(\frac{1}{2} (10\text{m})^2 \sin \left(\frac{360}{6} \right) \right) = 259.80\text{m}^2$$

$$A = A_C - A_H = \pi r^2 - 259.80\text{m}^2 = \pi(10)^2 - 259.80$$

$$A = 54.36\text{m}^2$$

239. Find the area of the regular pentagon whose side is 25m and apothem is 17.2m.

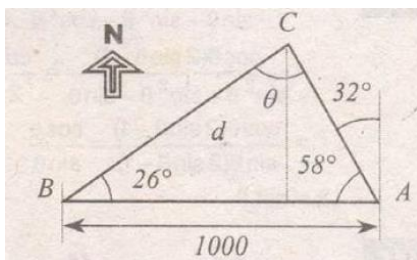
Solution:



$$A = 5A_T = 5 \left(\frac{1}{2} \right) bh = \frac{5}{2} (25\text{m})(17.2\text{m}) = 1075\text{m}^2$$

240. Points A and B 1000m apart are plotted on a straight highway running east and west. From A, the bearing of a tower C is 32° W of N and from the bearing of C is 26° N of E. Approximate the shortest distance of tower C to the highway.

Solution:



$$\theta = 180 - (26 + 58) = 96^\circ$$

By Sine Law:

$$\frac{\sin 96}{1000\text{m}} = \frac{\sin 58}{BC}; BC = 852.719\text{m}$$

$$\sin 26 = \frac{d}{BC}; d = 374\text{m}$$

241. The linear distance between -4 and 17 on the number line is?

Solution:

$$D = 17 + 4$$

$$D = 21$$

242. Find the distance between points A (4,-3) and B (-2, 5).

Solution:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(-2 - 4)^2 + (5 + 3)^2} = 10$$

243. If the distance between points (3, y) and (8, 7) is 13, then y is equal to?

Solution:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$(13)^2 = (8 - 3)^2 + (7 - y)^2$$

$$169 = 25 + (49 - 14y + y^2)$$

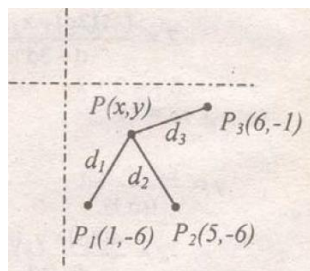
By factoring:

$$(y - 19)(y + 5) = 0$$

$$y = -5 \text{ \& } 19$$

244. Find the coordinates of a point equidistant from (1, -6), (5, -6) and (6, -1).

Solution:



$$d_1 = d_2$$

$$= \sqrt{(1 - x)^2 + (-6 - y)^2} = \sqrt{(5 - x)^2 + (-6 - y)^2}$$

$$= (1 - x)^2 + (-6 - y)^2 = (5 - x)^2 + (-6 - y)^2$$

$$= (1 - x)^2 = (5 - x)^2$$

$$= 1 - 2x + x^2 = 25 - 10x + x^2$$

$$x = 3$$

$$d_2 = d_3$$

$$\sqrt{(5 - x)^2 + (-6 - y)^2} = \sqrt{(6 - x)^2 + (-1 - y)^2}$$

$$= (5 - x)^2 + (-6 - y)^2 = (6 - x)^2 + (-1 - y)^2$$

Substitute $x = 3$:

$$y = -3$$

Thus, the point is at $(3, -3)$

245. If $(-2, -4)$ is the midpoint of $(6, -7)$ and (x, y) , then the values of x and y are?

Solution:

$$x_m = \frac{x_1 + x_2}{2}$$

$$-2 = \frac{6 + x}{2}$$

$$x = -10$$

$$y_m = \frac{y_1 + y_2}{2}$$

$$-4 = \frac{-7 + y}{2}$$

$$y = -1$$

246. Find the inclination of the line through $(-5, 3)$ and $(10, 7)$.

Solution:

Inclination means the angle or arctan of slope.

$$\tan \theta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7 - 3}{10 - (-5)} = 0.2666$$

$$\theta = 14.93^\circ$$

247. Find the angles formed by the lines $2x + y - 8 = 0$ and $x + 3y + 4 = 0$.

Solution:

By merely inspection,

$$2x + y - 8 = 0; y = -2x + 8$$

$$m_1 = -2$$

$$x + 3y + 4 = 0; y = -\frac{1}{3}x - \frac{4}{3}$$

$$m_2 = -\frac{1}{3}$$

$$\theta = \arctan \left[\frac{m_2 - m_1}{1 + m_1 m_2} \right] = \arctan \left[\frac{-\frac{1}{3} + 2}{1 + \left(-\frac{1}{3}\right)(-2)} \right]$$

$$\theta = 45^\circ$$

248. Convert 11.3099° to degree-feet.

Solution:

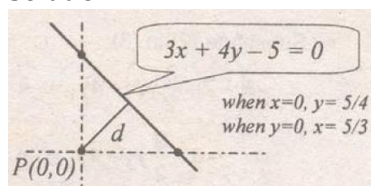
$$11 \text{ and } 0.3099 * \frac{60'}{1^\circ}$$

$$= 11 \text{ and } 18.594$$

$$= 11^\circ 19'$$

249. Find the distance of the line $3x + 4y = 5$ from the origin.

Solution:



$$d = \frac{Ax_1 + By_1 + C}{\pm \sqrt{A^2 + B^2}} = \frac{3(0) + 4(0) - 5}{\pm \sqrt{(3)^2 + (4)^2}} = -1$$

$$d = 1$$

250. The distance from the point $(2, 1)$ to the line $4x - 3y + 5 = 0$

Solution:

$$d = \frac{Ax_1 + By_1 + C}{\pm \sqrt{A^2 + B^2}} = d = \frac{4(2) + (-3)(1) + 5}{\pm \sqrt{(4)^2 + (-3)^2}} = -2$$

$$d = 2$$

251. Find the distance between the given lines $4x - 3y = 12$ and $4x - 3y = -8$

Solution:

In equation 1: $A = 4$, $B = -3$ and $C_1 = -12$

The C_2 of equation 2 is 8

$$d = \frac{C_1 - C_2}{\pm \sqrt{A^2 + B^2}} = \frac{-12 - 8}{\pm \sqrt{4^2 + (-3)^2}}$$

$$d = 4$$

252. What is the length of the line with a slope of $4/3$ from a point $(6, 4)$ to the y-axis?

Solution:

Using point slope form:

$$y - y_1 = m(x - x_1)$$

$$y - 4 = \frac{4}{3}(x - 6)$$

At y axis where $x = 0$,

$$y = -4$$

$$d = \sqrt{(6 - 0)^2 + (4 + 4)^2}$$

$$d = 10$$

253. In a Cartesian coordinates, the vertices of a triangle are defined by the following points: $(-2, 0)$, $(4, 0)$ and $(3, 3)$. What is the area?

Solution:

$$A = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} -2 & 4 & 3 \\ 0 & 0 & 3 \end{vmatrix} = -2$$

$$A = \frac{1}{2} [(-2)(0) + (4)(3) + (3)(0) - (0)(4) - (0)(3) - (3)(-2)]$$

$$A = 9$$

If the problem is a square then just add another row like $\begin{vmatrix} & & \\ & & \end{vmatrix}$

254. What are the coordinates of the center of the curve $x^2 + y^2 - 2x - 4y - 31 = 0$?

Solution:

$$x^2 + y^2 - 2x - 4y - 31 = 0$$

$$(x^2 - 2x + 1) + (y^2 - 4y + 4) = 0$$

$$(x - 1)^2 + (y - 2)^2 = 34$$

Thus, the coordinates are $(1, 2)$ and the radius is 6

255. Find the distance between the centers of the circles $x^2 + y^2 + 2x + 4y - 3 = 0$ and $x^2 + y^2 - 8x - 6y + 7 = 0$.

Solution:

The center of the first equation is $(-1, -2)$ and the second equation is $(4, 3)$

Thus, the distance is:

$$d = \sqrt{(4 + 1)^2 + (3 + 2)^2}$$

$$d = 7.07$$

256. The shortest distance from point A $(3, 8)$ to the circle $x^2 + y^2 + 4x - 6y = 12$ is equal to?

Solution:

By inspection: the center and radius of the equation of the circle is $C(-2, 3)$ and $r = 5$

$$d = \sqrt{(3 + 2)^2 + (8 - 3)^2} = 7.1$$

$$x = d - r$$

$$x = 2.1$$

257. The equation $x^2 + y^2 - 4x + 2y - 20 = 0$ describes?

Solution:

The equation is a circle with center at $C(2, -1)$ and radius $r = 5$

258. $3x^2 + 2x - 5y + 7 = 0$ Determine the curve:

Answer: parabola

259. The focus of the parabola $y^2 = 16x$ is at:

Solution:

The standard equation of a parabola is $y^2 = 4ax$

$$4a = 16$$

$$a = 4$$

Thus the focus is at $(4, 0)$

260. Where is the vertex of the parabola $x^2 = 4(y - 2)$?

Solution:

The point (h, k) means vertex:

The standard equation of this parabola is: $(x - h)^2 = 4(y - k)$

By inspection:

$h = 0$ and $k = 2$, thus the vertex is at $(0, 2)$

261. What is the length of the latus rectum of the curve $x^2 = 20y$?

Solution:

Latus rectum $= 4a$

$$4a = 20$$

262. What is the area enclosed by the curve $9x^2 + 25y^2 - 225 = 0$

Solution:

Note: the general equation of an ellipse is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Thus, the equation $9x^2 + 25y^2 - 225 = 0$

$$= \frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$= \frac{x^2}{(5)^2} + \frac{y^2}{(3)^2} = 1$$

By inspection:

$a = 5$ and $b = 3$

$$A = \pi ab = \pi(5)(3)$$

$$A = 47.12 \text{ units}$$

263. The length of the major and minor axis of an ellipse are 10m and 8m. Find the distance between the foci.

Solution:

Note: the distance between foci is $2c$

$$2a = 10 \quad 2b = 8$$

$$a = 5 \quad b = 4$$

$$c = \sqrt{a^2 - b^2} = \sqrt{(5)^2 - (4)^2} = 3$$

$$2c = 2(3) = 6$$

264. The equation $25x^2 + 16y^2 - 150x + 128y + 81 = 0$ has its center at?

Solution:

$$= 25x^2 - 150x + 16y^2 + 128y + 81 = 0$$

$$= 25(x^2 - 6x) + 16(y^2 + 8y) + 81 = 0$$

$$= 25(x^2 - 6x + 9) + 16(y^2 + 8y + 16) = (9)(25) + 16(16) - 81$$

$$= 25(x - 3)^2 + 16(y + 4)^2 = 400$$

$$= \frac{(x - 3)^2}{(4)^2} + \frac{(y + 4)^2}{(5)^2} = 1$$

By inspection:

$$(h, k) = (3, -4)$$

$a = 5$ and $b = 4$

265. Find the major axis of the ellipse $x^2 + 4y^2 - 2x - 8y + 1 = 0$.

Solution:

$$\begin{aligned} &= (x-1)^2 + 4(y-1)^2 = -1 + 4 + 1 \\ &= (x-1)^2 + 4(y-1)^2 = 4 \\ &= \frac{(x-1)^2}{(2)^2} + \frac{(y-1)^2}{1^2} = 1 \end{aligned}$$

By inspection, the length major axis is

$$2a = 2(2) = 4$$

266. The length of the latus rectum for the ellipse $\frac{x^2}{64} + \frac{y^2}{16} = 1$ is equal to:

Solution:

$$LR = \frac{2b^2}{a} = \frac{2(4)^2}{(8)} = 4$$

267. $4x^2 - y^2 = 16$ is an equation of?

Solution:

Note: Since the coefficient of x^2 and y^2 are opposite sign, then it is obvious that the equation is hyperbola.

$$\begin{aligned} &4x^2 - y^2 = 16 \\ &= \frac{x^2}{(2)^2} - \frac{y^2}{(4)^2} = 1 \end{aligned}$$

The equation is a Hyperbola

268. Find the distance between P (6, -2, -3) and Q (5, 1, -4).

Solution:

$$\begin{aligned} d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \\ d &= \sqrt{(5 - 6)^2 + (1 + 2)^2 + (-4 + 3)^2} \\ d &= \sqrt{11} \end{aligned}$$

269. What is the radius of the sphere center at the origin that passes the point 8, 1, 6?

Solution:

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} \\ r &= \sqrt{(8)^2 + (1)^2 + (6)^2} \\ r &= \sqrt{101} \end{aligned}$$

270. Evaluate: $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 + 3x - 4}$

Solution:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 + 3x - 4} = \lim_{x \rightarrow 1} \frac{(x+1)(x-1)}{(x+4)(x-1)} = \lim_{x \rightarrow 1} \frac{x+1}{x+4} = \frac{2}{5}$$

271. Differentiate: $y = \log_{10}(x^2 + 1)^2$

Solution:

Note:

$$d \log_{10} u = \log_{10} e \left(\frac{du}{u} \right)$$

Where:

$$u = (x^2 + 1)^2; du = 4x(x^2 + 1)$$

$$y' = \log_{10} e \left[\frac{4x(x^2 + 1)}{(x^2 + 1)^2} \right] = \frac{4x \log_{10} e}{x^2 + 1}$$

272. Differentiate: $\sqrt{(x^2 + 2)}$

Solution:

Note:

$$\frac{d}{dx} \sqrt{u} = \frac{du}{2\sqrt{u}}$$

$$\frac{d}{dx} \sqrt{(x^2 + 2)} = \frac{2x}{2\sqrt{(x^2 + 2)}} = \frac{x}{\sqrt{(x^2 + 2)}}$$

273. Find y' if $y = \sin^{-1}(\cos x)$

Solution:

Note:

$$d \sin^{-1} u = \frac{du}{\sqrt{1 - u^2}}$$

$$y' = \frac{-\sin x}{\sqrt{1 - \cos^2 x}}$$

Since: $\sin^2 x + \cos^2 x = 1$,

$$y' = \frac{-\sin x}{\sqrt{1 - \cos^2 x}} = \frac{-\sin x}{\sqrt{\sin^2 x}} = \frac{-\sin x}{\sin x} = -1$$

274. A farmer has enough money to build only 100m of fence. What are the dimensions of the field he can enclose the maximum area?

Solution:

perimeter: $2x + 2y = 100$; $y = 50 - x$

$A = xy = x(50 - x) = 50x - x^2$

$$\frac{dA}{dx} = 50 - 2x = 0$$

$x = 25$; $y = 25$

275. Find the minimum amount of tin sheet that can be made into a closed cylinder having a volume of 108in^3 .

Solution:

$V = \pi r^2 h$

$$108\text{in}^3 = \pi r^2 h; h = \frac{108\text{in}^3}{\pi r^2}$$

$$A = 2\pi r h + 2\pi r^2 = 2\pi r \left[\frac{108\text{in}^3}{\pi r^2} \right] + 2\pi r^2 = \frac{216}{r} + 2\pi r^2$$

$$\frac{dA}{dr} = -\frac{216}{r^2} + 4\pi r = 0$$

$$\frac{216}{r^2} = 4\pi r; r = 2.58\text{in}$$

$$h = \frac{108\text{in}^3}{\pi(2.58)^2} = 5.164\text{n}$$

$$A = 2\pi(2.58)(5.164) + 2\pi(2.58)^2 = 125.53\text{in}^2$$

276. The cost C of a product is a function of the quantity x of the product: $C(x) = x^2 - 4000x + 50$. find the quantity for which the cost is minimum.

Solution:

$C(x) = x^2 - 4000x + 50$

$$\frac{dC}{dx} = 2x - 4000; x = 2000 \text{ units}$$

277. A fencing is limited to 20ft length. What is the maximum rectangular area that can be fenced in using two perpendicular corner sides of an existing wall?

Solution:

Perimeter:

$$x + y = 20; y = 20 - x$$

Area:

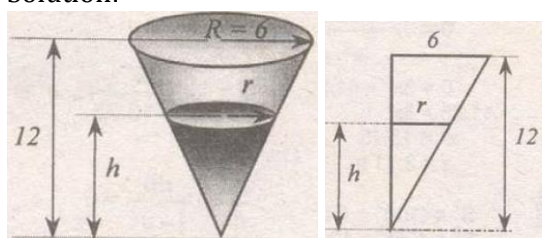
$$A = xy = x(20 - x) = 20x - x^2$$

$$\frac{dA}{dx} = 20 - 2x; x = 10\text{ft}; y = 10\text{ft}$$

$$A = (10\text{ft})(10\text{ft}) = 100\text{ft}^2$$

278. Water is flowing into a conical cistern at the rate of $8\text{m}^3/\text{min}$. If the height of the inverted cone is 12m and the radius of its circular opening is 6m. How fast is the water level rising when the water is 4m deep?

Solution:



$$V = \frac{1}{3} \pi r^2 h$$

By similar triangles:

$$\frac{6}{12} = \frac{r}{h}; r = \frac{h}{2}$$

$$V = \frac{1}{2} \pi \left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}$$

$$\frac{dV}{dt} = \frac{3\pi h^2}{12} \left(\frac{dh}{dt}\right)$$

$$\frac{dV}{dt} = \frac{8\text{m}^3}{\text{min}} \text{ \& } h = 4\text{m}$$

$$\frac{\left(\frac{8\text{m}^3}{\text{min}}\right)}{\frac{3\pi h^2}{12}} = \frac{dh}{dt} = \frac{\left(\frac{8\text{m}^3}{\text{min}}\right)}{\frac{3\pi(4)^2}{12}}$$

$$\frac{dh}{dt} = \frac{0.64\text{m}}{\text{min}}$$

279. Water is pouring into a swimming pool. After t hours, there are $t + \sqrt{t}$ gallons in the pool. At what rate is the water pouring into the pool when $t = 9$ hrs?

Solution:

$$Q = t + \sqrt{t}$$

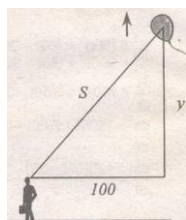
$$\frac{dQ}{dt} = 1 + \frac{1}{2\sqrt{t}}$$

When $t = 9$ hrs:

$$\frac{dQ}{dt} = 1 + \frac{1}{2\sqrt{t}} = 1 + \frac{1}{2\sqrt{9}} = \frac{7}{6} \text{ gph}$$

280. A balloon is released from the ground 100 meters from an observer. The balloon rises directly upward at the rate of 4 meters per second. How fast is the balloon receding from the observer 10 seconds later?

Solution:



$$S^2 = 100^2 + y^2$$

$$2S \left(\frac{dS}{dt} \right) = 2y \left(\frac{dy}{dt} \right)$$

$$\frac{dS}{dt} = \frac{y}{S} \left(\frac{dy}{dt} \right)$$

When $t = 10$ seconds,

$$y = V \cdot t = \left(\frac{4\text{m}}{\text{s}} \right) (10\text{s}) = 40\text{m}$$

$$S = \sqrt{100^2 + y^2} = \sqrt{100^2 + (40)^2} = 107.7\text{m}$$

$$\frac{dS}{dt} = \frac{y}{S} \left(\frac{dy}{dt} \right) = \frac{40}{107.7} \left(\frac{4\text{m}}{\text{s}} \right) = \frac{1.49\text{m}}{\text{s}}$$

281. What is the integral of $(3t - 1)^3 dt$?

Solution:

$$\int (3t - 1)^3 dt = \int \frac{u^3 du}{3} = \frac{1}{3} \left[\frac{(3t - 1)^4}{4} \right] + C = \frac{(3t - 1)^4}{12} + C$$

282. Evaluate the integral of $dx/(x + 2)$ from -6 to -10.

Solution:

$$\begin{aligned} \int_{-6}^{-10} \frac{dx}{x + 2} &= \ln(x + 2) = \ln(-10 + 2) - \ln(-6 + 2) = \ln(-8) - \ln(-4) \\ &= \ln \frac{-8}{-4} \\ &= \ln 2 \end{aligned}$$

283. Integrate $x \cos(2x^2 + 7) dx$.

Solution:

$$\text{Let } u = 2x^2 + 7$$

$$du = 4x dx$$

$$x dx = \frac{du}{4}$$

$$\int x \cos(2x^2 + 7) dx = \int \frac{\cos(u) du}{4} = \frac{1}{4} \sin(2x^2 + 7) + C$$

284. Integrate $(7x^3 + 4x^2) dx$

Solution:

$$\int (7x^3 + 4x^2) dx = \int 7x^3 dx + \int 4x^2 dx = \frac{7}{4} x^4 + \frac{4}{3} x^3 + C$$

285. Evaluate $\int_5^6 x(x - 5)^{12} dx$

Solution:

Note: $\int u dv = uv - \int v du$

$$u = x; du = dx; dv = (x-5)^{12} dx; v = \frac{(x-5)^{13}}{13}$$

$$\int_5^6 x(x-5)^{12} dx = \frac{x(x-5)^{13}}{13} - \frac{1}{13} \int (x-5)^{13} dx$$

$$= \frac{x(x-5)^{13}}{13} - \frac{1}{13} \left[\frac{(x-5)^{14}}{14} \right]$$

$$\left[\frac{(6)((6)-5)^{13}}{13} - \frac{1}{13} \left[\frac{((6)-5)^{14}}{14} \right] \right] - \left[\frac{(5)((5)-5)^{13}}{13} - \frac{1}{13} \left[\frac{((5)-5)^{14}}{14} \right] \right]$$

$$= 0.456$$

286. Evaluate $\int_0^1 \frac{x dx}{(x+1)^8}$

Solution:

$$\int_0^1 (x+1)^{-8} x dx$$

$$\text{Let: } u = x; dx = du; dv = (x+1)^{-8} dx; v = \frac{(x+1)^{-7}}{-7}$$

$$\int_0^1 (x+1)^{-8} x dx = uv - \int v du$$

$$= x \left[\frac{(x+1)^{-7}}{-7} \right] + \frac{1}{7} \int \frac{(x+1)^{-7} du}{-7}$$

$$= -\frac{x}{7(x+1)^7} + \frac{1}{7} \left[\frac{(x+1)^{-6}}{-6} \right]$$

$$= \left[-\frac{1}{7(1+1)^7} + \frac{1}{7} \left[\frac{(1+1)^{-6}}{-6} \right] \right] - \left[-\frac{0}{7(0+1)^7} + \frac{1}{7} \left[\frac{(0+1)^{-6}}{-6} \right] \right]$$

$$= -\frac{1}{672} + \frac{1}{42}$$

$$= \frac{5}{224} \text{ or } 0.22$$

287. Evaluate integral $\int \cos^2 y dy$

Solution:

$$\text{Note: } \cos^2 u = \frac{1}{2} \cos 2u; \text{ Type equation here.}$$

$$\int \cos^2 y dy = \frac{1}{2} \int 1 + \cos 2y dy = \frac{1}{2} \int dy + \frac{1}{2} \int \frac{\cos u du}{2}$$

$$= \frac{y}{2} + \frac{\sin 2y}{4} + C$$

288. Evaluate $\int \sqrt{1 - \cos x} dx$

Solution:

$$\text{Note: } \sin \frac{x}{2} = \sqrt{\frac{1 - \cos x}{2}} = \frac{\sqrt{1 - \cos x}}{\sqrt{2}}; \sqrt{2} \sin \frac{x}{2} = \sqrt{1 - \cos x}$$

$$\int \sqrt{1 - \cos x} dx = \int \sqrt{2} \sin \frac{x}{2} dx = 2\sqrt{2} \int \sin u du = 2\sqrt{2} \left[-\cos \frac{x}{2} \right] + C$$

$$= -2\sqrt{2} \cos \frac{x}{2} + C$$

289. Evaluate $\int_1^e \ln x dx$

Solution:

Let $u = \ln x$; $du = \frac{dx}{x}$; $dv = dx$; $v = x$

$$\begin{aligned}\int_1^e \ln x dx &= uv - \int v du = x \ln x - \int \frac{xdx}{x} = x \ln x - \int dx \\ &= x \ln x - x \\ &= [e \ln e - e] - [\ln 1 - 1] \\ &= e - e + 1 \\ &= 1\end{aligned}$$

290. Evaluate $\int_0^{\frac{\pi}{2}} \int_0^1 \int_0^2 z dz r^2 dr \sin u du$

Solution:

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \int_0^1 \int_0^2 z dz r^2 dr \sin u du &= \int_0^{\frac{\pi}{2}} \int_0^1 \frac{z^2}{2} = \int_0^{\frac{\pi}{2}} \int_0^1 2r^2 dr \sin u du = \int_0^{\frac{\pi}{2}} \frac{2}{3} r^3 = \frac{2}{3} \int_0^{\frac{\pi}{2}} \sin u du \\ &= \frac{2}{3} [-\cos u] = -\frac{2}{3} \cos \frac{\pi}{2} + \frac{2}{3} \cos 0 \\ &= \frac{2}{3}\end{aligned}$$

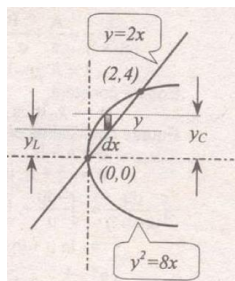
291. Evaluate $\int_0^2 \int_0^y (3x^2 + 9y^2) dx dy$

Solution:

$$\begin{aligned}\int_0^2 \int_0^y (3x^2 + 9y^2) dx dy &= \int_0^2 x^3 + 9y^2 x = \int_0^2 y^3 + 9y^3 dy = 10 \int_0^2 y^3 dy \\ &= \frac{10}{4} (2)^4 \\ &= 40\end{aligned}$$

292. Find the area of the region bounded by $y^2 = 8x$ and $y = 2x$

Solution:



$$\begin{aligned}y^2 &= 8x & y &= 2x \\ (2x)^2 &= 8x; x = 2; y = 4\end{aligned}$$

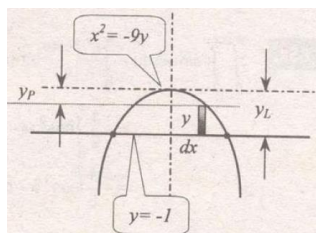
$$A = \int_0^2 y dx = \int_0^2 (\sqrt{8x} - 2x) dx = \frac{\sqrt{8}x^{\frac{3}{2}}}{\frac{3}{2}} - x^2 = \frac{4\sqrt{2}x^{\frac{3}{2}}}{3} - x^2$$

$$A = \left[\frac{4\sqrt{2}x^{\frac{3}{2}}}{3} - x^2 \right] - \left[\frac{4\sqrt{2}x^{\frac{3}{2}}}{3} - x^2 \right]$$

$$A = \frac{4}{3} \text{ or } 1.33 \text{ units}$$

293. What is the area bounded by the curve $x^2 = -9y$ and the line $y + 1 = 0$?

Solution:



$$x^2 = -9y = -9(-1) = 9$$

$$x = \pm 3$$

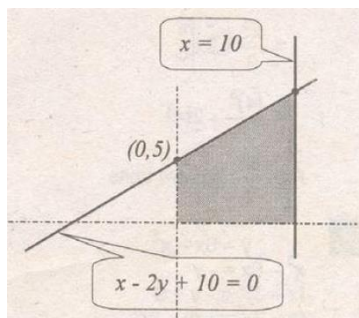
$$A = -2 \int_0^3 y dx = -2 \int_0^3 \left(-1 + \frac{x^2}{9}\right) dx = -2 \left[-x + \frac{x^3}{27}\right]$$

$$A = -2 \left[\left(-3 + \frac{3^3}{27}\right) - \left(-0 + \frac{0^3}{27}\right) \right]$$

$$A = 4 \text{ units}$$

294. Find the area bounded by the line $x - 2y + 10 = 0$, the x-axis, the y-axis and $x = 10$.

Solution:



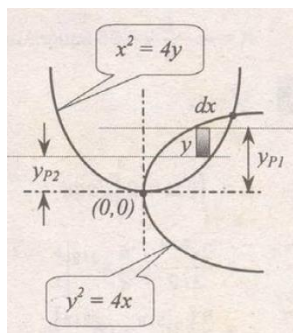
$$A = \int_0^{10} \left(\frac{x}{2} + 5\right) dx = \frac{x^2}{4} + 5x$$

$$A = \left[\frac{10^2}{4} + 5(10) \right] - \left[\frac{0^2}{4} + 5(0) \right]$$

$$A = 75 \text{ units}$$

295. What is the area bounded by the curve $y^2 = 4x$ and $x^2 = 4y$?

Solution:



$$x = \frac{y^2}{4}$$

$$\left(\frac{y^2}{4}\right)^2 = 4y; y = 4$$

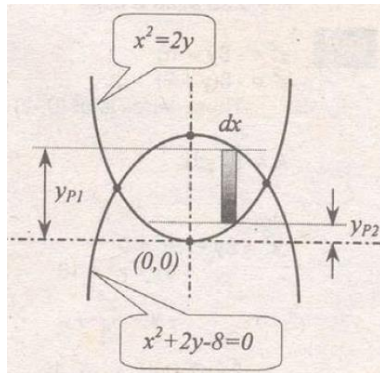
$$A = \int_0^4 \left(\sqrt{4x} - \frac{x^2}{4} \right) dx = \frac{4}{3} x^{\frac{3}{2}} - \frac{x^3}{12}$$

$$A = \left[\frac{4}{3} (4)^{\frac{3}{2}} - \frac{(4)^3}{12} \right] - \left[\frac{4}{3} (0)^{\frac{3}{2}} - \frac{(0)^3}{12} \right]$$

$$A = 5.33 \text{ or } \frac{16}{3} \text{ units}$$

296. Find the area bounded by the parabolas $x^2 - 2y = 0$ and $x^2 + 2y - 8 = 0$

Solution:



$x^2 = -2y + 8 = -2(y - 4) = (h, k) = (0, 4)$ vertex of this parabola

$$x^2 = 2y$$

$$2y = -2y + 8; y = 2$$

$$x^2 = 2y = \sqrt{2(2)} \pm 2$$

$$A = 2 \int_0^2 \left(\frac{8 - x^2}{2} - \frac{x^2}{2} \right) dx = 8x - \frac{1}{3}x^3 - \frac{1}{3}x^3 = 8x - \frac{2}{3}x^3$$

$$A = \left[8(2) - \frac{2}{3}(2)^3 \right] - \left[8(0) - \frac{2}{3}(0)^3 \right]$$

$$A = 10.67 \text{ or } \frac{32}{3} \text{ units}$$

297. Stream-flow in river X has a mean of $128\text{m}^3/\text{s}$ and a standard deviation of $52\text{m}^3/\text{s}$. If the sample data is normally distributed, the magnitude of stream-flow corresponding to a return period of 6.25 years is

Solution:

$$X = \frac{128\text{m}^3}{\text{s}} + \frac{52\text{m}^3}{\text{s}} = \frac{180\text{m}^3}{\text{s}}$$

298. A fully penetrating well installed in a confined aquifer with a transmissivity of $197\text{m}^2/\text{day}$ is pumped at a constant rate. If the drawdowns in two observation wells 8m and 22m from the pumped well were observed to be 1.42m and 0.44m, respectively. The steady state pump discharge is.

Solution:

$$Q = \frac{2\pi k t (h_i - h_w)}{\ln\left(\frac{r_i}{r_w}\right)} = \frac{2\pi \left(\frac{197\text{m}^2}{\text{day}}\right) (1.42\text{m} - 0.44)}{\ln\left(\frac{22}{8}\right)} = \frac{1199.12\text{m}^3}{\text{day}} \text{ or } 220\text{gpm}$$

299. The velocities at 0.2 and 0.8 of the depth of flow in a 1m wide rectangular open channel are 1.2m/s and 0.3m/s. If the flow depth is 0.8, the discharge is.

Solution:

$$Q = \left(\frac{\frac{1.2\text{m}}{\text{s}} + \frac{0.3\text{m}}{\text{s}}}{2} \right) (0.8\text{m}) = \frac{0.6\text{m}^3}{\text{s}}$$

300. Given the following direct runoff hydrograph in m^3/s taken in 1 hour interval in the mainstream of 50 square kilometer watershed: 0, 2.8, 13.8, 44.8, 77.8, 37.8, 12.8, 4.5, 1.8, 1.3, 0.9, 0.6, and 0.2. The equivalent depth of direct runoff is:

Solution:

$$X = \frac{\left(\frac{199.1\text{m}^3}{\text{s}} \right) (3600\text{s}) (1000\text{m})}{50,000,000\text{m}^2} = 14.30\text{mm}$$

301. A trapezoidal canal has a bottom width of 3m, a side slope of 1:1, a bottom slope of 0.02%, a roughness coefficient of 0.015. IF the discharge is $2\text{m}^3/\text{s}$, the normal depth is?

Give:

Bottom width (b) = 3m

Side slope = 1:1

Bottom slope = 0.02%

Roughness of coefficient (n) = 0.015

Discharge (Q) = $2\text{m}^3/\text{s}$

Solution:

$$AR^{\frac{2}{3}} = \frac{Qn}{\sqrt{S}} = \frac{\left(\frac{2\text{m}^3}{\text{s}} \right) (0.015)}{\sqrt{0.0002}} = 2.1$$

$$(by + zy^2) \left(\frac{by + zy^2}{b + 2y\sqrt{1 + z^2}} \right)^{\frac{2}{3}} = 2.1$$

$$(3y + y^2) \left(\frac{3y + y^2}{3 + 2y\sqrt{2}} \right)^{\frac{2}{3}} = 2.1$$

By trial and error or using calculator: substitute the choices given in the exam:

Let $y = 0.8\text{m}$

$$(3(0.7\text{m}) + (0.7\text{m})^2) \left(\frac{3(0.7\text{m}) + (0.7\text{m})^2}{3 + 2(0.7\text{m})\sqrt{2}} \right)^{\frac{2}{3}} = 2.1$$

Thus, the normal depth is $y = 0.8\text{m}$

302. The critical depth in a rectangular channel with a bottom width of 4m for discharge of $5\text{m}^3/\text{s}$.

Answer: (75% of the depth) = 0.54m

303. A hydro-electric plant having 50km^2 reservoir area and 100m head is used to generate power. The energy utilized by the consumers whose load is connected to the power plant during a five-hour period is $13.5 \times 10^6 \text{kW-hr}$. The overall generation efficiency is 75%. Find the fall in the height of water in the reservoir after 5hrs period.

Solution:

Total volume flow rate:

- 304.